31
Permanents

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The permanent is a matrix function introduced (independently) by Cauchy and Binet in 1812. At first sight it seems to be a simplified version of the determinant, but this impression is misleading. In some important respects the permanent is much less tractable than the determinant. Nonetheless, permanents have found a wide range of applications from pure combinatorics (e.g., counting problems involving permutations) right through to applied science (e.g., modeling subatomic particles). For further reading see [Min78], [Min83], [Min87], [CW05], and the references therein.

31.1 Basic Concepts

Definitions:

Let $A = [a_{ij}]$ be an $m \times n$ matrix over a commutative ring, $m \leq n$. Let S be the set of all injective functions from $\{1, 2, \ldots, m\}$ to $\{1, 2, \ldots, n\}$ (in particular, if $m = n$, then S is the symmetric group on $\{1, 2, \ldots, n\}$). The **permanent** of A is defined by

$$\operatorname{per}(A) = \sum_{\sigma \in S} \prod_{i=1}^{m} a_{i\sigma(i)}.$$

Two matrices A and B are **permutation equivalent** if there exist permutation matrices P and Q such that $B = PAQ$.

Facts:

For facts for which no specific reference is given and for background reading on the material in this subsection, see [Min78].

1. If A is any square matrix, then $\operatorname{per}(A) = \operatorname{per}(A^T)$.
2. Our definition implies that $\operatorname{per}(A) = 0$ for all $m \times n$ matrices A, where $m > n$. Some authors prefer to define $\operatorname{per}(A) = \operatorname{per}(A^T)$ in this case.
3. If A is any $m \times n$ matrix and P and Q are permutation matrices of respective orders m and n, then $\operatorname{per}(PAQ) = \operatorname{per}(A)$. That is, the permanent is invariant under permutation equivalence.
4. If A is any $m \times n$ matrix and c is any scalar, then $\operatorname{per}(cA) = c^m \operatorname{per}(A)$.
5. The permanent is a multilinear function of the rows. If $m = n$, it is also a multilinear function of the columns.
6. It is *not* in general true that $\operatorname{per}(AB) = \operatorname{per}(A)\operatorname{per}(B)$.
7. If M has block decomposition $M = \begin{bmatrix} A & \mathbf{0} \\ B & C \end{bmatrix}$, where either A or C is square, then $\operatorname{per}(M) = \operatorname{per}(A)\operatorname{per}(C)$.
8. Let A be an $m \times n$ matrix with $m \leq n$. Then $\operatorname{per}(A) = 0$ if A contains an $s \times (n - s + 1)$ submatrix of zeroes, for some $s \in \{1, 2, \ldots, m\}$.
9. (Laplace expansion) If A is an $m \times n$ matrix, $2 \leq m \leq n$, and $\alpha \in Q_{r,m}$, where $1 \leq r < m \leq n$ then
$$\operatorname{per}(A) = \sum_{\beta \in Q_{r,n}} \operatorname{per}\big(A[\alpha, \beta]\big) \operatorname{per}\big(A(\alpha, \beta)\big).$$
In particular, for any $i \in \{1, 2, \ldots, m\}$,
$$\operatorname{per}(A) = \sum_{j=1}^{n} a_{ij} \operatorname{per}\big(A(i, j)\big).$$
10. If A and B are $n \times n$ matrices and s and t are arbitrary scalars, then
$$\operatorname{per}(sA + tB) = \sum_{r=0}^{n} s^r t^{n-r} \sum_{\alpha, \beta \in Q_{r,n}} \operatorname{per}\big(A[\alpha, \beta]\big) \operatorname{per}\big(B(\alpha, \beta)\big)$$
(where we interpret the permanent of a 0×0 matrix to be 1).
11. (Binet–Cauchy) If A and B are $m \times n$ and $n \times m$ matrices, respectively, where $m \leq n$, then
$$\operatorname{per}(AB) = \sum_{\alpha} \frac{1}{\mu(\alpha)} \operatorname{per}\big(A[\{1, 2, \ldots, m\}, \alpha]\big) \operatorname{per}\big(B[\alpha, \{1, 2, \ldots, m\}]\big).$$
The sum is over all nondecreasing sequences α of m integers chosen from the set $\{1, 2, \ldots, n\}$ and $\mu(\alpha) = \alpha_1! \alpha_2! \cdots \alpha_n!$, where α_i denotes the number of occurrences of the integer i in α.
12. [MM62], [Bot67] Let F be a field and $m \geq 3$ an integer. Let T be a linear transformation for which $\operatorname{per}\big(T(X)\big) = \operatorname{per}(X)$ for all $X \in F^{m \times m}$. Then there exist permutation matrices P and Q and diagonal matrices D_1 and D_2 such that $\operatorname{per}(D_1 D_2) = 1$ and either $T(X) = D_1 P X Q D_2$ for all $X \in F^{m \times m}$ or $T(X) = D_1 P X^T Q D_2$ for all $X \in F^{m \times m}$.
13. (Alexandrov's inequality) Let $A = [a_{ij}] \in \mathbb{R}^{n \times n}$ and $1 \leq r < s \leq n$. If $a_{ij} \geq 0$ whenever $j \neq s$, then
$$(\operatorname{per}(A))^2 \geq \sum_{i=1}^{n} a_{ir} \operatorname{per}(A(i, s)) \sum_{i=1}^{n} a_{is} \operatorname{per}(A(i, r)). \tag{31.1}$$
Moreover, if $a_{ij} > 0$ whenever $j \neq s$, then equality holds in (31.1) iff there exists $c \in \mathbb{R}$ such that $a_{ir} = c a_{is}$ for all i.

14. If G is a balanced bipartite graph (meaning the two parts have equal size), then $\mathrm{per}(\mathcal{B}_G)$ counts perfect matchings (also known as 1-factors) in G.
15. If D is a directed graph, then $\mathrm{per}(\mathcal{A}_D)$ counts the cycle covers of D. A cycle cover is a set of disjoint cycles which include every vertex exactly once.

Examples:

1.

$$\mathrm{per}\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix} = 1\ \mathrm{per}\begin{bmatrix}5&6\\8&9\end{bmatrix} + 2\ \mathrm{per}\begin{bmatrix}4&6\\7&9\end{bmatrix} + 3\ \mathrm{per}\begin{bmatrix}4&5\\7&8\end{bmatrix}$$
$$= 1\cdot 5\cdot 9 + 1\cdot 6\cdot 8 + 2\cdot 4\cdot 9 + 2\cdot 6\cdot 7 + 3\cdot 4\cdot 8 + 3\cdot 5\cdot 7 = 450.$$

2. $\mathrm{per}\begin{bmatrix}0&2&3&4\\5&6&0&8\\9&0&0&12\end{bmatrix} = 2\cdot5\cdot12+2\cdot8\cdot9+3\cdot5\cdot12+3\cdot6\cdot9+3\cdot6\cdot12+3\cdot8\cdot9+4\cdot6\cdot9 = 1254.$
3. If $A = \begin{bmatrix}2&3\\2&-2\end{bmatrix}$ and $B = \begin{bmatrix}-4&-2\\1&-1\end{bmatrix}$, then $AB = \begin{bmatrix}-5&-7\\-10&-2\end{bmatrix}$. Hence, $80 = \mathrm{per}(AB) \neq \mathrm{per}(A)\,\mathrm{per}(B) = 2 \times 2 = 4$.
4. Below is a bipartite graph G (Figure 31.1) and its biadjacency matrix $\mathcal{B}_G$.

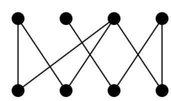
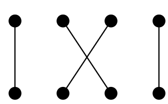


FIGURE 31.1

$$\begin{bmatrix}1&1&0&0\\0&0&1&0\\1&1&0&1\\0&0&1&1\end{bmatrix}$$

Now, $\mathrm{per}(\mathcal{B}_G) = 2$, which means that G has two perfect matchings (Figure 31.2 and Figure 31.3).



and

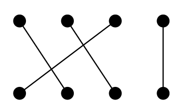


FIGURE 31.2 FIGURE 31.3

5. The matrix in the previous example can also be interpretted as $\mathcal{A}_D$ for the directed graph D (Figure 31.4). It has two cycle covers (Figure 31.5 and Figure 31.6).

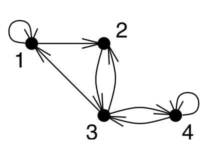


FIGURE 31.4

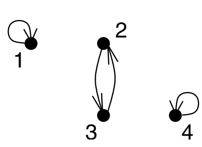


FIGURE 31.5

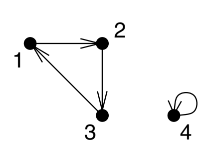


FIGURE 31.6

31.2 Doubly Stochastic Matrices

Facts:

For facts for which no specific reference is given and for background reading on the material in this subsection, see [Min78].

1. If $A \in \Omega_n$, then $\mathrm{per}(A) \leq 1$ with equality iff A is a permutation matrix.
2. [Ego81], [Fal81] If $A \in \Omega_n$ and $A \neq \frac{1}{n}J_n$, then $\mathrm{per}(A) > \mathrm{per}(\frac{1}{n}J_n) = n!/n^n$.

3. [Fri82] If $A \in \Omega_n$ and A has a $p \times q$ submatrix of zeros (where $p + q \leq n$), then

$$\text{per}(A) \geq \frac{(n-p)!}{(n-p)^{n-p}} \frac{(n-q)!}{(n-q)^{n-q}} \frac{(n-p-q)^{n-p-q}}{(n-p-q)!}.$$

   Equality is achieved by any matrix permutation equivalent to the matrix $A = [a_{ij}]$ defined by

$$a_{ij} = \begin{cases} 0 & \text{if } i \leq p \text{ and } j \leq q, \\ \frac{1}{n-q} & \text{if } i \leq p \text{ and } j > q, \\ \frac{1}{n-p} & \text{if } i > p \text{ and } j \leq q, \\ \frac{n-p-q}{(n-p)(n-q)} & \text{if } i > p \text{ and } j > q. \end{cases}$$

   If $p + q \neq n - 1$, then no other matrix achieves equality.
4. [BN66] If A is any $n \times n$ row substochastic matrix and $1 \leq r \leq n$, then $\text{per}(A) \leq m_r$, where m_r is the maximum permanent over all $r \times r$ submatrices of A.
5. [Min78, p. 41] If $A = [a_{ij}]$ is a fully indecomposable matrix, then the matrix $S = [s_{ij}]$ defined by $s_{ij} = a_{ij} \, \text{per}\big(A(i, j)\big) / \, \text{per}(A)$ is doubly stochastic and has the same zero pattern as A.

Examples:

1. The minimum value of the permanent in Ω_5 is 24/625, which is (uniquely) achieved by

$$\tfrac{1}{5} J_5 = \begin{bmatrix} 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \end{bmatrix}.$$

2. In the previous example, if we require that two specified entries in the same row must be zero, then the minimum value that the permanent can take is 1/24, which is (uniquely, up to permutation equivalence) achieved by

$$\begin{bmatrix} 0 & 0 & 1/3 & 1/3 & 1/3 \\ 1/4 & 1/4 & 1/6 & 1/6 & 1/6 \\ 1/4 & 1/4 & 1/6 & 1/6 & 1/6 \\ 1/4 & 1/4 & 1/6 & 1/6 & 1/6 \\ 1/4 & 1/4 & 1/6 & 1/6 & 1/6 \end{bmatrix}.$$

3. A nonnegative matrix of order $n \geq 3$ can have zero permanent even if we insist that each row and column sum is at least one. For example,

$$\text{per} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} = 0.$$

4. Suppose n identical balls are placed, one ball per bucket, in n labeled buckets on the back of a truck. When the truck goes over a bump the balls are flung into the air, but then fall back into the buckets. Suppose that the probability that the ball from bucket i lands in bucket j is p_{ij}. Then the matrix $P = [p_{ij}]$ is row stochastic and $\text{per}(P)$ is the probability that we end up with one ball in each bucket. That is, $\text{per}(P)$ is the *permanence* of the initial state.

31.3 Binary Matrices

Definitions:

A **binary matrix** is a matrix in which each entry is either 0 or 1.

Λ_n^k is the set of $n \times n$ binary matrices in which each row and column sum is k.

For an $m \times n$ binary matrix M the **complement** M^c is defined by $M^c = J_{mn} - M$.

A system of distinct representatives (**SDR**) for the finite sets $S_1, S_2, \ldots, S_n$ is a choice of $x_1, x_2, \ldots, x_n$ with the properties that $x_i \in S_i$ for each i and $x_i \neq x_j$ whenever $i \neq j$.

The **incidence matrix** for subsets $S_1, S_2, \ldots, S_m$ of a finite set $\{x_1, x_2, \ldots, x_n\}$ is the $m \times n$ binary matrix $M = [m_{ij}]$ in which $m_{ij} = 1$ iff $x_j \in S_i$.

$P_{(12\cdots n)}$ is the permutation matrix for the full cycle permutation $(12 \cdots n)$.

Facts:

For facts for which no specific reference is given and for background reading on the material in this section, see [Min78].

1. The number of SDRs for a set of sets with incidence matrix M is $\text{per}(M)$.
2. If $M \in \Lambda_n^k$, then $\frac{1}{k} M \in \Omega_n$, so the results of the previous subsection apply.
3. [Min78, p. 52] Let A be an $m \times n$ binary matrix where $m \leq n$. Suppose A has at least t positive entries in each row. If $t < m$ and $\text{per}(A) > 0$, then $\text{per}(A) \geq t!$. If $t \geq m$, then $\text{per}(A) \geq t!/(t-m)!$.
4. If $A \in \Lambda_n^k$, then there exist permutation matrices $P_1, P_2, \ldots, P_k$ such that
$$A = \sum_{i=1}^{k} P_i.$$
5. [Brè73], [Sch78] Let A be any $n \times n$ binary matrix with row sums $r_1, r_2, \ldots, r_n$. Then
$$\text{per}(A) \leq \prod_{i=1}^{n} (r_i!)^{1/r_i} \tag{31.2}$$
with equality iff A is permutation equivalent to a direct sum of square matrices each of which contains only 1s.
6. [MW98] If $m \geq 5$, then $(J_k \oplus J_k \oplus \cdots \oplus J_k)^c$ (where there are m copies of J_k) maximizes the permanent in Λ_{mk}^{mk-k}. The result is not true for $m = 3$.
7. [Wan99b] For each $k \geq 1$ there exists N such that for all $n \geq N$ a matrix M maximizes the permanent in Λ_n^k iff $M \oplus J_k$ maximizes the permanent in Λ_{n+k}^k.
8. [Wan03] If $n = tk + r$ with $0 \leq r < k \leq n$, then $k!^t r! \leq \max_{A \in \Lambda_n^k} \text{per}(A) \leq (k!)^{n/k}$.
9. [Wan03] If $k = o(n)$ as $n \to \infty$, then $\left(\max_{A \in \Lambda_n^k} \text{per}(A) \right)^{1/n} \sim (k!)^{1/k}$.
10. [GM90] Suppose $0 \leq k = O(n^{1-\delta})$ for a constant $\delta > 0$ as $n \to \infty$. Then
$$\text{per}(A) = n! \left(\frac{n-k}{n} \right)^n \exp\left(\frac{k}{2n} + \frac{3k^2 - k}{6n^2} + \frac{2k^3 - k}{4n^3} \right.$$
$$\left. + \frac{15k^4 + 70k^3 - 105k^2 + 32k}{60n^4} + \frac{z}{n^4} + \frac{2z(2k-1)}{n^5} + O\left(\frac{k^5}{n^5} \right) \right)$$
for all $A \in \Lambda_n^{n-k}$, where z denotes the number of 2×2 submatrices of A that contain only zeros. In particular, if $0 \leq k = O(n^{1-\delta})$ for a constant $\delta > 0$ as $n \to \infty$, then $\text{per}(A)$ is asymptotically equal to $n!(1 - k/n)^n$ for all $A \in \Lambda_n^{n-k}$.
11. [Wan06] If $2 \leq k \leq n$ as $n \to \infty$, then
$$\left(\min_{A \in \Lambda_n^k} \text{per}(A) \right)^{1/n} \sim \frac{(k-1)^{k-1}}{k^{k-2}}.$$

12. [BN65] For any integers $k \geq 1$ and $n \geq \log_2 k + 1$ there exists a binary matrix A of order n such that $\text{per}(A) = k$.
13. The permanent of a square binary matrix counts **permutations with restricted positions**. This means that for each point being permuted there is some set of allowable images, while other images are forbidden.

Examples:

1. If $D_n = I_n^c$, then

$$\text{per}(D_n) = n!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \cdots + (-1)^n \frac{1}{n!}\right)$$

is the number of **derangements** of n things, that is, the number of permutations of n points that leave no point in its original place. The 4^{th} derangement number is

$$\text{per}(D_4) = \text{per}\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} = 9.$$

2. The number of ways that n married couples can sit around a circular table with men and women alternating and so that nobody sits next to their spouse is $2\,n!\,M_n$, where M_n is known as the n-th **menagé number** and is given by

$$M_n = \text{per}\left((I_n + P_{(12\cdots n)})^c\right) = \sum_{r=0}^{n} (-1)^r \frac{2n}{2n-r}\binom{2n-r}{r}(n-r)!.$$

The 5^{th} menagé number is $M_5 = \text{per}\begin{bmatrix} 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix} = 13.$

3. SDRs are important for many combinatorial problems. For example, a $k \times n$ **Latin rectangle** (where $k \leq n$) is a $k \times n$ matrix in which n symbols occur in such a way that each symbol occurs exactly once in each row and at most once in each column. The number of extensions of a given $k \times n$ Latin rectangle R to a $(k+1) \times n$ Latin rectangle is the number of SDRs of the sets $S_1, S_2, \ldots, S_n$ defined so that S_i consists of the symbols not yet used in column i of R.
4. [CW05] For $n \leq 11$ the minimum values of $\text{per}(A)$ for $A \in \Lambda_n^k$ are as follows:

k	$n=2$	3	4	5	6	7	8	9	10	11
1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2
3	–	6	9	12	17	24	33	42	60	83
4	–	–	24	44	80	144	248	440	764	1316
5	–	–	–	120	265	578	1249	2681	5713	12105
6	–	–	–	–	720	1854	4738	12000	30240	75510
7	–	–	–	–	–	5040	14833	43386	126117	364503
8	–	–	–	–	–	–	40320	133496	439792	1441788
9	–	–	–	–	–	–	–	362880	1334961	4890740
10	–	–	–	–	–	–	–	–	3628800	14684570
11	–	–	–	–	–	–	–	–	–	39916800

5. [MW98] For $n \leq 11$ the maximum values of $\text{per}(A)$ for $A \in \Lambda_n^k$ are as follows:

k	$n = 2$	3	4	5	6	7	8	9	10	11
1	1	1	1	1	1	1	1	1	1	1
2	2	2	4	4	8	8	16	16	32	32
3	–	6	9	13	36	54	81	216	324	486
4	–	–	24	44	82	148	576	1056	1968	3608
5	–	–	–	120	265	580	1313	2916	14400	31800
6	–	–	–	–	720	1854	4752	12108	32826	86400
7	–	–	–	–	–	5040	14833	43424	127044	373208
8	–	–	–	–	–	–	40320	133496	440192	1448640
9	–	–	–	–	–	–	–	362880	1334961	4893072
10	–	–	–	–	–	–	–	–	3628800	14684570
11	–	–	–	–	–	–	–	–	–	39916800

31.4 Nonnegative Matrices

Definitions:

Δ_n^k is the set of $n \times n$ matrices of nonnegative integers in which each row and column sum is k.

Facts:

1. [Min78, p. 33] Let A be an $m \times n$ nonnegative matrix with $m \leq n$. Then $\text{per}(A) = 0$ iff A contains an $s \times (n - s + 1)$ submatrix of zeros, for some $s \in \{1, 2, \ldots, m\}$.
2. [Min78, p. 38] Let A be a nonnegative matrix of order $n \geq 2$. Then A is fully indecomposable iff $\text{per}\big(A(i, j)\big) > 0$ for all i, j.
3. [Sch98] $\left(\frac{(k-1)^{k-1}}{k^{k-2}}\right)^n \leq \min\limits_{A \in \Delta_n^k} \text{per}(A) \leq \frac{k^{2n}}{\binom{kn}{n}}$.
4. [Min83] It is conjectured that $\min\limits_{A \in \Delta_n^k} \text{per}(A) = \min\limits_{A \in \Lambda_n^k} \text{per}(A)$.
5. [Sou03] Let Γ denote the gamma function and let A be a nonnegative matrix of order n. In row i of A, let m_i and r_i denote, respectively, the largest entry and the total of the entries. Then,
$$\text{per}(A) \leq \prod_{i=1}^{n} m_i \left(\Gamma\left(\frac{r_i}{m_i} + 1\right)\right)^{m_i / r_i}. \tag{31.3}$$
6. [Min78, p. 62] Let A be a nonnegative matrix of order n. Define s_i to be the sum of the i smallest entries in row i of A. Similarly, define S_i to be the sum of the i largest entries in row i of A. Then
$$\prod_{i=1}^{n} s_i \leq \text{per}(A) \leq \prod_{i=1}^{n} S_i.$$
7. [Gib72] If A is nonnegative and π is a root of $\text{per}(zI - A)$, then $|\pi| \leq \rho(A)$.

Examples:

1. [BB67] If A is row substochastic, the roots of $\text{per}(zI - A) = 0$ satisfy $|z| \leq 1$.
2. If Soules' bound (31.3) is applied to matrices in Λ_n^k it reduces to Brègman's bound (31.2).

31.5 (± 1)-Matrices

Facts:

1. [KS83] If A is a (± 1)-matrix of order n, then $\operatorname{per}(A)$ is divisible by $2^{n-\lfloor \log_2(n+1) \rfloor}$.
2. [Wan74] If H is an $n \times n$ Hadamard matrix, then $|\operatorname{per}(H)| \leq |\det(H)| = n^{n/2}$.
3. [KS83], [Wan05] There is no solution to $|\operatorname{per}(A)| = |\det(A)|$ among the nonsingular (± 1)-matrices of order n when $n \in \{2, 3, 4\}$ or $n = 2^k - 1$ for $k \geq 2$, but there are solutions when $n \in \{5, \ldots, 20\} \setminus \{7, 15\}$.
4. [Wan05] There exists a (± 1)-matrix A of order n satisfying $\operatorname{per}(A) = 0$ iff $n + 1$ is not a power of 2.

Examples:

1. The 11×11 matrix $A = [a_{ij}]$ defined by

$$a_{ij} = \begin{cases} -1 & \text{if } j - i \in \{1, 2, 3, 5, 7, 10\}, \\ +1 & \text{otherwise,} \end{cases}$$

satisfies $\operatorname{per}(A) = 0$. No smaller (± 1)-matrix of order $n \equiv 3 \bmod 4$ has this property.

2. The following matrix has per $=$ det $= 16$, and is the smallest example (excluding the trivial case of order 1) for which per $=$ det among ± 1-matrices.

$$\begin{bmatrix} +1 & +1 & +1 & +1 & +1 \\ +1 & -1 & -1 & +1 & +1 \\ +1 & -1 & +1 & +1 & +1 \\ +1 & +1 & +1 & -1 & -1 \\ +1 & +1 & +1 & -1 & +1 \end{bmatrix}.$$

31.6 Matrices over $\mathbb{C}$

Facts:

1. If $A = [a_{ij}] \in \mathbb{C}^{m \times n}$ and $B = [b_{ij}] \in \mathbb{R}^{m \times n}$ satisfy $b_{ij} = |a_{ij}|$, then $|\operatorname{per}(A)| \leq \operatorname{per}(B)$.
2. If $A \in \mathbb{C}^{n \times n}$, then $\operatorname{per}(\overline{A}) = \overline{\operatorname{per}(A)} = \operatorname{per}(A^*)$.
3. [Min78, p. 113] If $A \in \mathbb{C}^{n \times n}$ is normal with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, then

$$|\operatorname{per}(A)| \leq \frac{1}{n} \sum_{i=1}^{n} |\lambda_i|^n.$$

4. [Min78, p. 115] Let $A \in \mathbb{C}^{n \times n}$ and let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of AA^*. Then

$$|\operatorname{per}(A)|^2 \leq \frac{1}{n} \sum_{i=1}^{n} \lambda_i^n.$$

5. [Lie66] If $A = \begin{bmatrix} B & C \\ C^* & D \end{bmatrix} \in \mathrm{PD}_n$, then $\operatorname{per}(A) \geq \operatorname{per}(B) \operatorname{per}(D)$.
6. [JP87] Suppose $\alpha \subseteq \beta \subseteq \{1, 2, \ldots, n\}$. Then for any $A \in \mathrm{PD}_n$,

$$\det(A[\beta, \beta]) \operatorname{per}(A(\beta, \beta)) \leq \det(A[\alpha, \alpha]) \operatorname{per}(A(\alpha, \alpha)).$$

(We interpret det or per of a 0×0 matrix to be 1.)

7. [MN62] If $A \in \mathbb{C}^{m\times n}$ and $B \in \mathbb{C}^{n\times m}$, then

$$|\operatorname{per}(AB)|^2 \leq \operatorname{per}(AA^*)\operatorname{per}(B^*B).$$

8. [Bre59] If $A = [a_{ij}] \in \mathbb{C}^{n\times n}$ satisfies $|a_{ii}| > \sum_{j\neq i} |a_{ij}|$ for each i, then $\operatorname{per}(A) \neq 0$.

Examples:

1. If U is a unitary matrix, then $|\operatorname{per}(U)| \leq 1$.

31.7 Subpermanents

Definitions:

The k-th **subpermanent sum**, $\operatorname{per}_k(A)$ of an $m \times n$ matrix A, is defined to be the sum of the permanents of all order k submatrices of A. That is,

$$\operatorname{per}_k(A) = \sum_{\substack{\alpha\in Q_{k,m}\\ \beta\in Q_{k,n}}} \operatorname{per}(A[\alpha,\beta]).$$

By convention, we define $\operatorname{per}_0(A) = 1$.

Facts:

For facts for which no specific reference is given and for background reading on the material in this subsection see [Min78].

1. For each k, per_k is invariant under permutation equivalence and transposition.
2. If A is any $m \times n$ matrix and c is any scalar, then $\operatorname{per}_k(cA) = c^k \operatorname{per}_k(A)$.
3. [BN66] If A is any $n \times n$ row substochastic matrix and $1 \leq r \leq n$, then $\operatorname{per}_r(A) \leq \binom{n}{r}$.
4. [Fri82] $\operatorname{per}_k(A) \geq \operatorname{per}_k(\frac{1}{n}J_n)$ for every $A \in \Omega_n$ and integer k.
5. [Wan03] If $A \in \Lambda_n^k$ and $i \leq k$, then $\operatorname{per}_i(A) \geq \binom{n}{i}\dfrac{k!}{(k-i)!}$.
6. [Wan03] For $1 \leq i, k \leq n$ and $A \in \Lambda_n^k$,

$$\binom{kn}{i} - kn(k-1)\binom{kn-2}{i-2} \leq \operatorname{per}_i(A) \leq \binom{kn}{i}.$$

7. [Wan03] For $A \in \Lambda_n^k$, let $\xi_i = \left(\operatorname{per}_i(A)/\binom{n}{i}\right)^{1/i}$. Then

$$\frac{(k-1)^{k-1}}{k^{k-2}} \leq \xi_n \leq \xi_{n-1} \leq \cdots \leq \xi_1 = k.$$

8. [Nij76], [HLP52, p. 104] Let A be a nonnegative $m \times n$ matrix with $\operatorname{per}(A) \neq 0$. For $1 \leq i \leq m-1$,

$$\frac{\operatorname{per}_i(A)}{\operatorname{per}_{i-1}(A)} \geq \frac{(i+1)(m-i+1)}{i(m-i)}\frac{\operatorname{per}_{i+1}(A)}{\operatorname{per}_i(A)} > \frac{\operatorname{per}_{i+1}(A)}{\operatorname{per}_i(A)}.$$

9. [Wan99b] For $A \in \Lambda_n^k$,

$$\frac{\operatorname{per}_i(A)}{\operatorname{per}_{i+1}(A)} \geq \frac{i+1}{(n-i)^2}.$$

10. [Wan99a] Let $k \geq 0$ be an integer. There exists no polynomial $p_k(n)$ such that for all n and $A \in \Lambda_n^3$,

$$\frac{\operatorname{per}_{n-k-1}(A)}{\operatorname{per}_{n-k}(A)} \leq p_k(n).$$

11. If G is a bipartite graph, then $\operatorname{per}_k(\mathcal{B}_G)$ counts the k-matchings in G. A k-matching in G is a set of k edges in G such that no two edges share a vertex.

Examples:

1. For any matrix A the sum of the entries in A is $\operatorname{per}_1(A)$.
2. Any $m \times n$ matrix A has $\operatorname{per}_m(A) = \operatorname{per}(A)$.
3. $\operatorname{per}_k(J_n) = k!\binom{n}{k}^2$.
4. [Wan99a] Let $A \in \Lambda_n^k$ have s submatrices which are copies of J_2. Then
   - $\operatorname{per}_1(A) = kn$.
   - $\operatorname{per}_2(A) = \frac{1}{2}kn(kn - 2k + 1)$.
   - $\operatorname{per}_3(A) = \frac{1}{6}kn(k^2n^2 - 6k^2n + 3kn + 10k^2 - 12k + 4)$.
   - $\operatorname{per}_4(A) = \frac{1}{24}kn\big(k^3n^3 - 12k^3n^2 + 6k^2n^2 + 52k^3n - 60k^2n + 19kn - 84k^3 + 168k^2 - 120k + 30\big) + s$.
   - $\operatorname{per}_5(A) = \frac{1}{120}kn\big[k^4n^4 - 20k^4n^3 + 10k^3n^3 + 160k^4n^2 - 180k^3n^2 + 55k^2n^2 - 620k^4n + 1180k^3n - 800k^2n + 190kn + 1008k^4 - 2880k^3 + 3240k^2 - 1680k + 336\big] + (nk - 8k + 8)s$.
5. The subpermanent sums are also known as **rook numbers** since they count the number of placements of rooks in mutually nonattacking positions on a chessboard. Let A be a binary matrix in which each 1 denotes a permitted position on the board and each 0 denotes a forbidden position for a rook. Then $\operatorname{per}_i(A)$ is the number of placements of i rooks on permitted squares so that no two rooks occupy the same row or column. For example, the number of ways of putting 4 nonattacking rooks on the white squares of a standard chessboard is

$$\operatorname{per}_4\begin{bmatrix} 1&0&1&0&1&0&1&0\\ 0&1&0&1&0&1&0&1\\ 1&0&1&0&1&0&1&0\\ 0&1&0&1&0&1&0&1\\ 1&0&1&0&1&0&1&0\\ 0&1&0&1&0&1&0&1\\ 1&0&1&0&1&0&1&0\\ 0&1&0&1&0&1&0&1 \end{bmatrix} = 8304.$$

31.8 Rook Polynomials

Definitions:

Let A be an $m \times n$ binary matrix. The polynomials $\rho_1(A,x) = \sum_{i=0}^{m} \operatorname{per}_i(A)x^i$ and $\rho_2(A,x) = \sum_{i=0}^{m}(-1)^i \operatorname{per}_i(A)x^{m-i}$ are both called **rook polynomials** because they are generating functions for the rook numbers.

Let $\ell_k(x)$ be the k^{th} Laguerre polynomial, normalized to be monic. That is,

$$\ell_k(x) = (-1)^k k! \sum_{i=0}^{k} \binom{k}{i} \frac{(-x)^i}{i!}.$$

Facts:

For facts for which no specific reference is given and for background reading on the material in this section, see [Rio58].

1. When A is an $m \times n$ binary matrix, $\rho_1(A,x) = (-x)^m \rho_2(A, -\frac{1}{x})$.
2. If $A = B \oplus C$, then $\rho_1(A,x) = \rho_1(B,x)\rho_1(C,x)$ and $\rho_2(A,x) = \rho_2(B,x)\rho_2(C,x)$.
3. [HL72], [Nij76] For any nonnegative matrix A, all the roots of $\rho_1(A,x)$ and $\rho_2(A,x)$ are real.
4. [HL72] For $2 \leq k \leq n$ and $A \in \Lambda_n^k$, the roots of $\rho_1(A,x)$ are less than $-1/(4k-4)$, while the roots of $\rho_2(A,x)$ lie in the interval $(0, 4k-4)$.
5. [JR80], [God81] For a square binary matrix A, with complement A^c,
$$\operatorname{per}(A) = \int_0^\infty e^{-x} \rho_2(A^c, x)\, \mathrm{dx}.$$
6. [God81] For an $n \times n$ binary matrix M,
$$\rho_2(M,x) = \sum_{i=0}^{n} \operatorname{per}_i(M^c)\ell_{n-i}(x).$$
7. [Sze75, p. 100] For any j,k let $\delta_{j,k}$ denote the Kronecker delta. Then
$$\int_0^\infty e^{-x} \ell_j(x)\ell_k(x)\, \mathrm{dx} = \delta_{j,k} k!^2.$$

Examples:

1. $\rho_1(P,x) = (x+1)^n$ and $\rho_2(P,x) = (x-1)^n$ for a permutation matrix $P \in \Lambda_n^1$.
2. $\rho_2(J_k, x) = \ell_k(x)$.
3. Let $C_n = I_n + P_{(12\cdots n)}$. Then
$$\rho_1(C_n, x) = \sum_{i=0}^{n} \frac{2n}{2n-i}\binom{2n-i}{i} x^i$$
and $\rho_2(C_n, 4x^2) = 2T_{2n}(x)$, where $T_n(x)$ is the n^{th} Chebyshev polynomial of the first kind.
4. $\rho_2(I_n^c, x) = \sum_{i=0}^{n} \binom{n}{i} \ell_i(x)$ for any integer $n \geq 1$.
5. The ideas of this chapter allow a quick computation of the permanent of many matrices that are built from blocks of ones by recursive use of direct sums and complementation. For example,
$$\begin{aligned}\operatorname{per}\left(\left(J_k \oplus I_{k+1}^c\right)^c\right) &= \int_0^\infty e^{-x} \rho_2(J_k \oplus I_{k+1}^c)\, \mathrm{dx} = \int_0^\infty e^{-x} \rho_2(J_k)\rho_2(I_{k+1}^c)\, \mathrm{dx} \\ &= \int_0^\infty e^{-x} \ell_k(x) \sum_{i=0}^{k+1} \binom{k+1}{i} \ell_i(x)\, \mathrm{dx} = \binom{k+1}{k} k!^2 = (k+1)!\,k!.\end{aligned}$$

31.9 Evaluating Permanents

Facts:

For facts for which no specific reference is given and for background reading on the material in this subsection, see [Min78].

1. Since the permanent is *not* invariant under elementary row operations, it cannot be calculated by Gaussian elimination.
2. (Ryser's formula) If $A = [a_{ij}]$ is any $n \times n$ matrix,
$$\operatorname{per}(A) = \sum_{r=1}^{n} (-1)^r \sum_{\alpha \in Q_{r,n}} \prod_{i=1}^{n} \sum_{j \in \alpha} a_{ij}.$$

3. [NW78] A straightforward implementation of Ryser's formula has time complexity $\Theta(n^2 2^n)$. By enumerating the α in Gray Code order (i.e., by choosing an ordering in which any two consecutive α's differ by a single entry), Nijenhuis/Wilf improved this to $\Theta(n2^n)$. They cut the execution time by a further factor of two by exploiting the relationship between the term corresponding to α and that corresponding to $\{1, 2, \ldots, n\} \setminus \alpha$. This second savings is not always desirable when calculating permanents of integer matrices, since it introduces fractions.

Ryser/Nijenhuis/Wilf (RNW) Algorithm for calculating per(A) for $A = [a_{ij}]$ of order n.

```
p := −1;
for i from 1 to n do
      x_i := a_in − (1/2) ∑_{j=1}^n a_ij;
      p := p ∗ x_i;
      g_i := 0;
s := −1;
for k from 2 to 2^(n−1) do
      if k is even then
            j := 1;
      else
            j := 2;
            while g_(j−1) = 0 do
                  j := j + 1;
      z := 1 − 2 ∗ g_j;
      g_j := 1 − g_j;
      s := −s;
      t := s;
      for i from 1 to n do
            x_i := x_i + z ∗ a_ij;
            t := t ∗ x_i;
      p := p + t;
return(2(−1)^n p)
```

4. For sufficiently sparse matrices, a simple enumeration of nonzero diagonals by backtracking, or a recursive Laplace expansion, will be faster than RNW.
5. A hybrid approach is to use Laplace expansion to expand any rows or columns that have very few nonzero entries, then employ RNW.
6. [DL92] The calculation of the permanent is a $\#P$-complete problem. This is still true if attention is restricted to matrices in Λ_n^3. So, it is extremely unlikely that a polynomial time algorithm for calculating permanents exists.
7. [Lub90] As a result of the above, much work has been done on approximation algorithms for permanents.

31.10 Connections between Determinants and Permanents

Definitions:

For any partition λ of n let χ_λ denote the irreducible character of the symmetric group S_n associated with λ by the standard bijection (see Section 68.6 or [Mac95, p. 114]) between partitions and irreducible characters.

Let ε_n be the identity in S_n.

The matrix function f_λ defined by

$$f_\lambda(M) = \frac{1}{\chi_\lambda(\varepsilon_n)} \sum_{\sigma \in S_n} \chi_\lambda(\sigma) \prod_{i=1}^{n} m_{i\sigma(i)},$$

for each $M = [m_{ij}] \in \mathbb{C}^{n\times n}$, is called a **normalized immanant** (without the factor of $1/\chi_\lambda(\varepsilon_n)$ it is an **immanant**).

The partial order $\lhd$ is defined on the set of partitions of an integer n by stating that $\lambda \lhd \mu$ means that $f_\lambda(H) \leq f_\mu(H)$ for all $H \in \mathrm{PD}_n$.

Facts:

For facts for which no specific reference is given and for background reading on the material in this subsection, see [Mer97].

1. If $\lambda = (n)$, then χ_λ is the principal/trivial character and f_λ is the permanent.
2. If $\lambda = (1^n)$, then χ_λ is the alternating character and f_λ is the determinant.
3. [Sch18] $f_\lambda(H)$ is a nonnegative real number for all $H \in \mathrm{PD}_n$ and all λ.
4. [MM61] For $n \geq 3$ there is no linear transformation T such that $\mathrm{per}(A) = \det\big(T(A)\big)$ for every $A \in \mathbb{R}^{n\times n}$. In particular, there is no way of affixing minus signs to some entries that will convert the permanent into a determinant.
5. [Lev73] For all sufficiently large n there exists a fully indecomposable matrix $A \in \Lambda_n^3$ such that $\mathrm{per}(A) = \det(A)$.
6. [Sch18], [Mar64] If $A = [a_{ij}] \in \mathrm{PD}_n$, then $\det(A) \leq \prod_{i=1}^{n} a_{ii} \leq \mathrm{per}(A)$. Equality holds iff A is diagonal or A has a zero row/column.
7. [Sch18] For arbitrary λ, if $A \in \mathrm{PD}_n$, then $f_\lambda(A) \geq \det(A)$. In other words $(1^n) \lhd \lambda$ for all partitions λ of n.
8. [Hey88] The hook immanants are linearly ordered between det and per. That is, $(1^n) \lhd (2, 1^{n-2}) \lhd (3, 1^{n-3}) \lhd \cdots \lhd (n-1, 1) \lhd (n)$.
9. A special case of the **permanental dominance conjecture** asserts that $\lambda \lhd (n)$ for all partitions λ of n. This has been proven only for special cases, which include (a) $n \leq 13$, (b) partitions with no more than three parts which exceed 2, and (c) the hook immanants mentioned above.
10. For two partitions λ, μ of n to satisfy $\lambda \lhd \mu$, it is necessary but not sufficient that μ majorizes λ.

Examples:

1. Although $\mu = (3, 1)$ majorizes $\lambda = (2, 2)$, neither $\lambda \lhd \mu$ nor $\mu \lhd \lambda$. This is demonstrated by taking $A = J_2 \oplus J_2$ and $B = J_1 \oplus J_3$ and noting that $f_\lambda(A) = 2 > \frac{4}{3} = f_\mu(A)$ but $f_\lambda(B) = 0 < 2 = f_\mu(B)$.

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32

D-Optimal Matrices

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An $m \times n$ matrix W whose entries are all 1 or -1 is called a (± 1)-design matrix; if the entries of W are 0 or 1, then W is a $(0, 1)$-design matrix. Each design matrix corresponds to a weighing design. That is, a scheme for estimating the weights of n objects in m weighings. Since the weights of n objects cannot be estimated in fewer than n weighings, we consider only those pairs (m, n) with $m \geq n$. The rows of W encode a two-pan or one-pan weighing design with n objects $x_1, \ldots, x_n$ being weighed in m weighings. If $W \in \{\pm 1\}^{m \times n}$, an entry of 1 in the (i, j)-th position of W indicates that object x_j is put in the right pan in the i-th weighing while an entry of -1 means that x_j is placed in the left pan. If $W \in \{0, 1\}^{m \times n}$, an entry of 1 in the (i, j)-th position indicates that object x_j is included in the i-th weighing while an entry of 0 means that the object is not included. In the presence of errors for the scale, we can expect only to find estimators $\hat{w}_1, \ldots, \hat{w}_n$ for the actual weights $w_1, \ldots, w_n$ of the objects. We want to choose a weighing design that is optimal with respect to some condition, an idea going back to Hotelling [Hot44] and Mood [Moo46]. See also [HS79] and [Slo79]. Under certain assumptions on the error of the scale, we can express optimality conditions in terms of $W^T W$ (see [Puk93]). The value of det $W^T W$ is inversely proportional to the volume of the confidence region of the estimators of the weights of the objects. Thus, matrices for which det $W^T W$ is large correspond to weighing designs that are desirable.

32.1 Introduction

This section includes basic definitions; facts and examples can be found in the following sections.

Definitions:

A matrix $W \in \{\pm 1\}^{m \times n}$ is a (± 1)-**design matrix**; $W \in \{0, 1\}^{m \times n}$ is a $(0, 1)$-**design matrix**.

A matrix $W \in \{\pm 1\}^{m \times n}$ (respectively, $W \in \{0, 1\}^{m \times n}$) is called **D-optimal** if det $W^T W$ is maximal over all matrices in $\{\pm 1\}^{m \times n}$ (respectively, $\{0, 1\}^{m \times n}$).

$\alpha(m, n) = \max\{\det W^T W | W \in \{\pm 1\}^{m \times n}\}$.

$\beta(m, n) = \max\{\det W^T W | W \in \{0, 1\}^{m \times n}\}$.

We write $\alpha(n)$ and $\beta(n)$ for $\alpha(n, n)$ and $\beta(n, n)$.

32.2 The (± 1) and $(0, 1)$ Square Case

Definitions:

For a matrix $V \in \{0, 1\}^{(n-1)\times(n-1)}$, define

$$W_V = \left[\begin{array}{c|ccc} 1 & 1 & \cdots & 1 \\ \hline -1 & & & \\ \vdots & & 2V - J_{n-1} & \\ -1 & & & \end{array}\right] \in \{\pm 1\}^{n\times n}.$$

For a matrix $W \in \{\pm 1\}^{n\times n}$, define $V_W \in \{0, 1\}^{(n-1)\times(n-1)}$ by

$$V_W = \frac{1}{2}\left(W(1) + J_{n-1}\right),$$

where $W(1)$ is obtained from W by deleting the first row and column.

A **signature matrix** is a ± 1 diagonal matrix.

A **Hadamard matrix** of order n is a matrix $H_n \in \{\pm 1\}^{n\times n}$ with $H_n H_n^T = nI_n$.

A **2-design** with parameters (v, k, λ) (also called a (v, k, λ)-**design**) is a collection of k-subsets B_i, called **blocks**, of a finite set X with cardinality v, such that each 2-subset of X is contained in exactly λ blocks.

The (± 1)-**incidence matrix** $W = (w_{ij})$ of a 2-design is a matrix whose rows are indexed by the elements x_i of X and whose columns are indexed by the blocks B_j. The entry $w_{ij} = -1$ if $x_i \in B_j$ and $w_{ij} = +1$ otherwise.

Facts:

1. For any $W \in \{\pm 1\}^{n\times n}$, there exist signature matrices S_1, S_2 such that for $W' = (S_1 W S_2)$, $W'_{1j} = 1$ for $j = 1, \ldots, n$ and $W'_{i1} = -1$ for $i = 2, \ldots, n$. W is D-optimal if and only if W' is D-optimal.
2. $\det W_V = 2^{n-1} \det V$.
3. [Wil46] The (± 1) square case in dimension n is related to the $(0, 1)$ square case in dimension $n - 1$ by the previous two facts, and $\alpha(n) = 4^{n-1}\beta(n - 1)$. Facts are stated here for $\alpha(n)$ only, since the facts for $\beta(n)$ can be easily derived from these.
4. [Had93], [CRC96], [BJL93], [WPS72] Hadamard matrices
   - A necessary condition for the existence of a Hadamard matrix of order n is $n = 1, n = 2$, or $n \equiv 0 \pmod 4$.
   - Let H_m and H_n be two Hadamard matrices of orders m and n, respectively. Then $H_m \otimes H_n$ is a Hadamard matrix of order mn.
   - There exist infinitely many values n for which a Hadamard matrix H_n exists.
   - It is conjectured that for all $n = 4k$ there exists a Hadamard matrix H_n.
   - The smallest n for which the existence of a Hadamard matrix is in question (at the time of the writing of this chapter) is $n = 668$.
5. [Had93], [CRC96], [BJL93], [WPS72] D-optimal (± 1)-matrices: the case $n = 4k$
   - $\alpha(4k) \leq (4k)^{4k}$.
   - A necessary and sufficient condition for equality to occur in this case is the existence of a Hadamard matrix of order n.
6. [Bar33], [Woj64], [Ehl64a], [Coh67], [Neu97] D-optimal (± 1)-matrices: the case $n = 4k + 1$
   - $\alpha(4k + 1) \leq (8k + 1)\,(4k)^{4k}$.
   - For equality to occur in this case, it is necessary and sufficient that $8k + 1$ is the square of an integer and that there exists a matrix $W \in \{\pm 1\}^{m\times n}$ with $W^T W = (n - 1)I_n + J_n$.

- Equality occurs for infinitely many values of $n = 4k + 1$. A.E. Brouwer ([Bro83]) constructed an infinite family of 2-designs with parameters $(n = 2q^2 + 2q + 1, q^2, (q^2 - q)/2)$. The (± 1)-incidence matrix W_n of such a design satisfies $W_n^T W_n = (n-1)I_n + J_n$.
- The results in [MK82] and [CMK87] provide upper bounds, which are stronger than $(8k + 1)(4k)^{4k}$ in case $8k + 1$ is not the square of an integer.

7. [Ehl64a], [Woj64] D-optimal (± 1)-matrices: the case $n = 4k + 2$
   - $\alpha(4k+2) \leq (8k+2)^2\,(4k)^{4k}$.
   - For equality to occur in this case, it is necessary that $n - 1$ is the sum of two squares and that there exists a matrix $W_n \in \{\pm 1\}^{n\times n}$ such that

$$W_n^T W_n = \begin{bmatrix} (n-2)I_{\frac{n}{2}} + 2J_{\frac{n}{2}} & 0_n \\ 0_n & (n-2)I_{\frac{n}{2}} + 2J_{\frac{n}{2}} \end{bmatrix}. \tag{32.1}$$

   - It is conjectured that the bound is attained whenever this is the case.
   - The bound is attained infinitely often.
   - If $n - 1$ is a square and there exists a matrix $W_{\frac{n}{2}} \in \{\pm 1\}^{\frac{n}{2}\times\frac{n}{2}}$ such that $W_{\frac{n}{2}}^T W_{\frac{n}{2}} = \frac{n-2}{2} I_{\frac{n}{2}} + J_{\frac{n}{2}}$, then construct the matrix $W_n = W_{\frac{n}{2}} \otimes H_2$ where $H_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. Then $W_n \in \{\pm 1\}^{n\times n}$ satisfies Equation (32.1) and attains the bound. Such a matrix $W_{\frac{n}{2}}$ exists if $\frac{n}{2} = 2q^2 + 2q + 1$.
8. [Ehl64b] D-optimal (± 1)-matrices: The case $n = 4k + 3$

$$\alpha(4k+3) \leq (4k)^{4k+3-s}(4k+4r)^u(4k+4+4r)^v \left(1 - \frac{ur}{4k+4r} - \frac{v(r+1)}{4k+4+4r}\right),$$

   where $s = 5$ for $k = 1$, $s = 5$ or 6 for $k = 2$, $s = 6$ for $3 \leq k \leq 14$, and $s = 7$ for $k \geq 15$ and where $r = \lfloor (4k+3)/s \rfloor$, $4k + 3 = rs + v$, and $u = s - v$.
   - This case is the least well understood of the four.
   - For equality to occur for $n \geq 63$, it is necessary that $n = 112j^2 \pm 28j + 7$ and that there exists a matrix $W_n \in \{\pm 1\}^{n\times n}$ with

$$W_n^T W_n = I_7 \otimes \left[(n-3)I_{\frac{n}{7}} + 4J_{\frac{n}{7}}\right] - J_n.$$

   - However, it is not known if this bound is attainable for any $n \geq 63$.
   - The best lower bound seems to be the one in [NR97]. In [NR97], an infinite family of matrices is constructed whose determinants attain about 37% of the bound above.
9. See [OS] for (± 1)-matrices with largest known determinant for $n \leq 103$.

Examples:

1. The following matrices are Hadamard matrices in $\{\pm 1\}^{4\times 4}$ and $\{\pm 1\}^{12\times 12}$:

$$H_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

$$H_{12} = \begin{bmatrix} 1 & 1 & 1 & 1 & -1 & -1 & -1 & 1 & 1 & 1 & -1 & -1 \\ 1 & 1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & -1 & -1 & 1 & 1 & 1 & -1 & -1 & -1 & 1 \\ 1 & -1 & -1 & 1 & -1 & -1 & 1 & -1 & -1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 & 1 & -1 & -1 & 1 & -1 & -1 & -1 & -1 \\ -1 & -1 & 1 & -1 & -1 & 1 & -1 & -1 & 1 & -1 & -1 & -1 \\ -1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & -1 & -1 & -1 & 1 & -1 & 1 \\ 1 & 1 & -1 & -1 & -1 & 1 & -1 & -1 & -1 & 1 & 1 & -1 \\ 1 & -1 & -1 & -1 & -1 & -1 & -1 & 1 & 1 & -1 & 1 & 1 \\ -1 & 1 & -1 & -1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 & 1 \\ -1 & -1 & 1 & -1 & -1 & -1 & 1 & 1 & -1 & 1 & 1 & -1 \end{bmatrix}$$

$H_4 H_4^T = 4I_4$ and $H_{12} H_{12}^T = 12I_{12}$.

2. Let $A = J_5 - 2I_5$. Then $A^T A = 4I_5 + J_5$ and $\det(A^T A)$ achieves the upper bound in Fact 6.
3. Let

$$W = \begin{bmatrix} A & A \\ A & -A \end{bmatrix} = A \otimes H_2 \in \{\pm 1\}^{10\times 10},$$

where $A = J_5 - 2I_5$. Then

$$W^T W = \begin{bmatrix} 8I_5 + 2J_5 & 0 \\ 0 & 8I_5 + 2J_5 \end{bmatrix},$$

and, hence, $\det(W^T W)$ achieves the upper bound in Fact 7.

4. To obtain the upper bound in Fact 6 for $n = 13$, let $V \in \{0,1\}^{13\times 13}$ be the $(0,1)$ line-point incidence matrix for a projective plane of order 3. Then $V^T V = 3I_{13} + J_{13}$ and the matrix $W = J_{13} - 2V \in \{\pm 1\}^{13\times 13}$ satisfies $W^T W = 12I_{13} + J_{13}$ and its determinant attains the upper bound.
5. For $n \geq 11$ and $n \equiv 3 \pmod 4$, no (± 1)-matrix is known to have a determinant that equals the upper bound in Fact 8. However, the following matrix $W \in \{\pm 1\}^{15\times 15}$, which is listed in [OS], satisfies

$$\det(W^T W) = 174755568785817600,$$

which is about 94% of the upper bound 185454889323724800 in Fact 8 with $k = 3$, $s = 6$.

$$W = \begin{bmatrix} -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 & 1 & -1 & 1 & 1 & 1 & 1 & -1 \\ -1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 & 1 & 1 & 1 & 1 & -1 & 1 & 1 \\ -1 & -1 & -1 & 1 & 1 & -1 & 1 & 1 & -1 & 1 & 1 & -1 & 1 & 1 & 1 \\ -1 & 1 & 1 & -1 & -1 & 1 & 1 & 1 & -1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & 1 & -1 & 1 & -1 & -1 & 1 & 1 & 1 & -1 & 1 & -1 & -1 & -1 & -1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & 1 & 1 & 1 & 1 & 1 & -1 & 1 & 1 & 1 & 1 & -1 & -1 & 1 & -1 \\ -1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & 1 & -1 & 1 & -1 & 1 & 1 & 1 \\ 1 & 1 & 1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 & -1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 & -1 & -1 & -1 & 1 & -1 & -1 & 1 & 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 & 1 & -1 \\ 1 & 1 & -1 & -1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & -1 & 1 & 1 & -1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & 1 & 1 & -1 \\ -1 & 1 & 1 & 1 & 1 & -1 & 1 & -1 & -1 & -1 & -1 & 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & 1 & 1 & 1 & -1 & 1 \end{bmatrix}.$$

32.3 The (± 1) Nonsquare Case

Facts:

1. [Had93] $\alpha(4k, n) \leq (4k)^n$.
2. [Pay74] $\alpha(4k+1, n) \leq (4k+n)(4k)^{n-1}$.
3. [Pay74]

$$\alpha(4k+2, n) \leq \begin{cases} (4k+n)^2(4k)^{n-2}, & \text{if } n \text{ is even,} \\ (4k+n+1)(4k+n-1)(4k)^{n-2}, & \text{if } n \text{ is odd.} \end{cases} \tag{32.2}$$

4. [GK80b] If $m = 4k - 1 \geq 2n - 5$, then $\alpha(m, n) \leq (4k - n)(4k)^{n-1}$.

Examples:

1. If $m = 4k$, equality can be achieved by taking W_0 to be the matrix consisting of any n columns of a Hadamard matrix of order $4k$. Then $W_0^T W_0 = 4kI_n$ and, hence, $\det W_0^T W_0 = \alpha(4k, n)$.
2. If $m = 4k + 1$, adjoin a row of all 1s to W_0 and call the new matrix W_1. We have $W_1 \in \{\pm 1\}^{(4k+1)\times n}$ and $W_1^T W_1 = mI_n + J$. Hence, $\det W_1^T W_1 = \alpha(4k+1, n)$.
3. If $m = 4k + 2$, let $r_1 = (1, 1, \ldots, 1)$ and $r_2 = (1, \ldots, 1, -1, \ldots, -1)$ be n-tuples with $r_1 \cdot r_2 = 0$, if n is even, and $r_1 \cdot r_2 = 1$, if n is odd. Adjoin rows r_1 and r_2 to W_0. Call the resulting matrix W_2. Then $W_2 \in \{\pm 1\}^{(4k+2)\times n}$ and

$$W_2^T W_2 = \begin{bmatrix} 4kI_l + 2J_l & 0_l \\ 0_l & 4kI_l + 2J_l \end{bmatrix} \quad \text{if } n = 2l \text{ is even}$$

and

$$W_2^T W_2 = \begin{bmatrix} 4kI_{l+1} + 2J_{l+1} & 0_{l+1,l} \\ 0_{l,l+1} & 4kI_l + 2J_l \end{bmatrix} \quad \text{if } n = 2l+1 \text{ is odd.}$$

   Thus, $\det W_2^T W_2 = \alpha(4k+2, n)$.
4. If $m = 4k - 1$, we may assume without loss of generality that the first row of W_0 is an all 1s row. Remove this first row of W_0 and call that matrix W_{-1}. Note that $W_{-1} \in \{\pm 1\}^{(4k-1)\times n}$ and that $W_{-1}^T W_{-1} = 4kI_n - J_n$. Hence, $\det W_{-1}^T W_{-1} = \alpha(4k-1, n)$.
5. It is not necessary to have a Hadamard matrix H_m of order m. All we require is the existence of a matrix $W \in \{\pm 1\}^{4k\times n}$ with $W^T W = mI_n$. See [GK80a] for details.
6. Upper bounds on $\alpha(4k-1, n)$ when $m = 4k - 1 \leq 2n - 5$ are given in [GK80b], [KF84], [SS91].

32.4 The $(0, 1)$ Nonsquare Case: Regular D-Optimal Matrices

Definitions:

Let W be a $(0,1)$-design matrix in $\{0,1\}^{m\times n}$. For n odd, W is **balanced** if every row of W has exactly $(n+1)/2$ ones; for n even, W is **balanced** if every row of W has exactly $n/2$ ones or exactly $(n+2)/2$ ones.

A design matrix $W \in \{0,1\}^{m\times n}$ is **regular** if it is balanced and $W^T W = t(I + J)$ for some integer t.

Facts:

1. [HKL96] If n is odd, then $\beta(m,n) \leq (n+1)\left(\frac{(n+1)m}{4n}\right)^n$, with equality if and only if W is regular.
2. [NWZ97] If n is even, then $\beta(m,n) \leq (n+1)\left(\frac{(n+2)m}{4(n+1)}\right)^n$, with equality if and only if W is regular.
3. [NWZ98a] A regular matrix exists in $\{0,1\}^{m\times n}$ only if

$$\begin{aligned} 2(n+1) \text{ divides } m \text{ for } n &\equiv 0 \pmod 4, \\ 2n \text{ divides } m \text{ for } n &\equiv 1 \pmod 4, \\ n+1 \text{ divides } m \text{ for } n &\equiv 2 \pmod 4, \\ n \text{ divides } m \text{ for } n &\equiv 3 \pmod 4. \end{aligned}$$

4. [NWZ98a] If $n = 4t-1$ and $H \in \{0,1\}^{m\times n}$ is the incidence matrix for a $(4t-1, 2t-1, t-1)$-design, then $W = J - H$ is a regular D-optimal matrix and $[\overbrace{W^T, \cdots, W^T}^{k}]^T$ is a regular D-optimal matrix in $\{0,1\}^{kn\times n}$. Let W_1 be the matrix obtained by deleting any column from W. Then $[\overbrace{W_1^T, \cdots, W_1^T}^{k}]^T$ is a regular D-optimal matrix in $\{0,1\}^{kn\times(n-1)}$.
5. [NWZ98a] If $n = 4t+1$ is a power of a prime integer, then a D-optimal regular matrix $W_2 \in \{0,1\}^{2n\times n}$ exists. Let $W_3 \in \{0,1\}^{2n\times(n-1)}$ be the matrix obtained by deleting any column from W_2. Then $[\overbrace{W_2^T, \cdots, W_2^T}^{k}]^T$ and $[\overbrace{W_3^T, \cdots, W_3^T}^{k}]^T$ are regular D-optimal matrices.

Examples:

1. Let $n = 4$ and $m = 10$. The following matrix is balanced and regular:

$$W^T = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}, \qquad W^TW = \begin{bmatrix} 6 & 3 & 3 & 3 \\ 3 & 6 & 3 & 3 \\ 3 & 3 & 6 & 3 \\ 3 & 3 & 3 & 6 \end{bmatrix}.$$

The inequality in Fact 2 is attained at W:

$$\beta(10,4) = 5\left(\frac{6\cdot 10}{4\cdot 5}\right)^4 = 405 = \det(W^TW).$$

Thus W is D-optimal.

2. A regular matrix exists for the case $n = 9, m = 18$. (Fact 3, where $n \equiv 1 \pmod 4$ and $2n = 18$ divides $m = 18$.) The regular matrix is constructed from the Galois field $GF(9)$ with nine elements. Choosing $GF(9) = \mathbb{Z}_3/(x^2+1)$, the element $\theta = x+2$ of order 8, generates the nonzero elements in $GF(9)$. The nonzero quadratic residues of $GF(9)$ are $Q = \{1, \theta^2, \theta^4, \theta^6\}$. Define $K_1 \in \{0,1\}^{9\times 9}$ by

$$(K_1)_{\rho,\tau} = \begin{cases} 1, & \text{if } \tau \in \rho + Q, \\ 0, & \text{if otherwise,} \end{cases}$$

where the rows and columns ρ, τ are indexed by $\{0, 1, \theta, \theta^2, \ldots, \theta^7\}$. Then

$$K_1 = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}.$$

Define K_2 in the same way but with the nonzero quadratic nonresidues $R = \{\theta, \theta^3, \theta^5.\theta^7\}$ in place of Q. Then the matrix $[K_1, K_2] \in \{0, 1\}^{9\times 18}$ satisfies

$$K_1 K_1^T + K_2 K_2^T = 5I_9 + 3J_9.$$

Let $W = [J_9 - K_1, J_9 - K_2]$. Then W is a D-optimal regular design matrix: $WW^T = 5(I_9 + J_9)$.

3. Let $t = 2$ and $n = 7$. The following matrix H is the incidence matrix for a $(7, 3, 1)$-design:

$$H = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \end{bmatrix}.$$

Then (Fact 4) $W = J_7 - H$ is a regular D-optimal matrix in $\{0, 1\}^{7\times 7}$ and $[W^T, \ldots, W^T]^T$ is a regular D-optimal matrix in $\{0, 1\}^{7k\times 7}$.

32.5 The $(0, 1)$ Nonsquare Case: Nonregular D-Optimal Matrices

It is clear from Fact 3 in section 32.4 that for most pairs (m, n), no regular D-optimal matrix exists. For example, if $n = 7$, then the only values of m for which a regular D-optimal matrix exists are $m = 7t$. Thus, for $m = 7t + r$, with $0 \le r \le 6$, a D-optimal matrix cannot be regular unless $r = 0$. The only values of n for which $\beta(m, n)$ is known for all values of m are $n = 2, 3, 4, 5, 6$.

Facts:

1. [HKL96] $n = 2$, $m = 3t + r$ with $r = 0, 1, 2$:

$$\beta(m, 2) = \begin{cases} 3t^2 \quad , & \text{for } r = 0 \\ 3t^2 + 2t, & \text{for } r = 1 \end{cases}.$$

2. [HKL96] $n = 3$, $m = 3t + r$ with $r = 0, 1, 2$:

$$\beta(m, 3) = 4t^{3-r}(t + 1)^r.$$

3. [NWZ98b] $n = 4$, $m = 10t + r$ with $0 \le r \le 9$:

$$\beta(m,4) = \begin{cases} 405\,t^4 & , \quad \text{for } r{=}0 \\ 81\,t^3\,(2+5\,t) & , \quad \text{for } r{=}1 \\ 3\,t\,(1+3\,t)^2\,(2+15\,t) & , \quad \text{for } r{=}2 \\ 3\,t\,(1+3\,t)^2\,(8+15\,t) & , \quad \text{for } r{=}3 \\ 3\,(1+3\,t)^3\,(3+5\,t) & , \quad \text{for } r{=}4 \\ (1+3\,t)^2\,\left(19+60\,t+45\,t^2\right) & , \quad \text{for } r{=}5 \\ 3\,(2+3\,t)^3\,(2+5\,t) & , \quad \text{for } r{=}6 \\ 3\,(1+t)\,(2+3\,t)^2\,(7+15\,t) & , \quad \text{for } r{=}7 \\ 3\,(1+t)\,(2+3\,t)^2\,(13+15\,t) & , \quad \text{for } r{=}8 \\ 81\,(1+t)^3\,(3+5\,t) & , \quad \text{for } r{=}9 \end{cases}.$$

4. [NWZ98b] In the case $n = 5$, all D-optimal matrices are balanced except when $m = 5, 6, 7, 8, 15, 16, 17, 27$. For $m = 10t + r$ with $0 \le r \le 9$ and m not equal to any of the exceptional values, we have

$$\beta(m,5) = \begin{cases} 1458\,t^5 & , \quad \text{for } r{=}0 \\ 729\,t^4\,(1+2\,t) & , \quad \text{for } r{=}1 \\ 162\,t^3\,(1+3\,t)\,(2+3\,t) & , \quad \text{for } r{=}2 \\ 27\,t^2\,(1+3\,t)^2\,(5+6\,t) & , \quad \text{for } r{=}3 \\ 54\,t\,(1+t)\,(1+3\,t)^3 & , \quad \text{for } r{=}4 \\ 9\,(1+t)\,(1+3\,t)^2\,\left(1+15\,t+18\,t^2\right) & , \quad \text{for } r{=}5 \\ 54\,(1+t)^2\,(1+3\,t)^3 & , \quad \text{for } r{=}6 \\ 27\,(1+t)^2\,(1+3\,t)^2\,(5+6\,t) & , \quad \text{for } r{=}7 \\ 162\,(1+t)^3\,(1+3\,t)\,(2+3\,t) & , \quad \text{for } r{=}8 \\ 729\,(1+t)^4\,(1+2\,t) & , \quad \text{for } r{=}8 \end{cases}.$$

The values of $\beta(m,5)$ for the eight exceptional values of m are

$$\begin{array}{llll} \beta(5,5) = 25 & \beta(6,5) = 64 & \beta(7,5) = 192 & \beta(8,5) = 384 \\ \beta(15,5) = 9880 & \beta(16,5) = 13975 & \beta(17,5) = 19500 & \beta(27,5) = 202752. \end{array}$$

Each of these is greater than the value of the corresponding polynomial above. For example, if $m = 16$ so that $t = 1$ and $r = 6$, then $54\,(1+t)^2\,(1+3\,t)^3 = 13824$, which is less than 13975.

5. [NWZ00] For $n = 6$, all D-optimal matrices are balanced except when $m = 6, 8, 9, 13$. For $m = 7t + r$ with $0 \le r \le 6$ and $m \ne 6, 8, 9, 13$:

$$\beta(m,6) = \begin{cases} 448\,t^6 & , \quad \text{for } r{=}0 \\ 16\,t^4\,(1+2\,t)\,(5+14\,t) & , \quad \text{for } r{=}1 \\ 4\,t^2\,(1+2\,t)^3\,(3+14\,t) & , \quad \text{for } r{=}2 \\ (1+2\,t)^5\,(1+14\,t) & , \quad \text{for } r{=}3 \\ (1+2\,t)^5\,(13+14\,t) & , \quad \text{for } r{=}4 \\ 4\,(1+t)^2\,(1+2\,t)^3\,(11+14\,t) & , \quad \text{for } r{=}5 \\ 16\,(1+t)^4\,(1+2\,t)\,(9+14\,t) & , \quad \text{for } r{=}6 \end{cases}$$

The values of $\beta(6, m)$ for the four exceptional values of m are

$$\beta(6,6) = 81 \qquad \beta(8,6) = 832 \qquad \beta(9,6) = 1620 \qquad \beta(13,6) = 16512.$$

As in the case for $n = 5$, the values of $\beta(m, 6)$ exceed the value of the corresponding polynomial.

Examples:

1. Design matrices are exhibited for the above cases in the sources listed. For example, if $n = 4$, let

$$W_0 = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}^T,$$

v_i be the ith row of W_0. If $r = 3$, then the matrix $[\overbrace{W_0^T, \cdots, W_0^T}^{t}, v_8^T, v_9^T, v_{10}^T]^T$ is a D-optimal matrix in $\{0, 1\}^{(4t+3)\times 4}$.

32.6 The $(0, 1)$ Nonsquare Case: Large m

Facts:

1. [NWZ98a] For each value of n, all D-optimal matrices in $\{0, 1\}^{m\times n}$ are balanced for sufficiently large values of m.
2. [NW02], [AFN03] In addition to the values $n = 2, 3, 4, 5, 6$, for which $\beta(m, n)$ is known for all m, the only other values of n for which $\beta(m, n)$ is known for all sufficiently large values of m are $n = 7, 11, 15, 19, 23, 27$.

Examples:

1. [NW02]

$$\beta(7t + r, 7) = 2^{10} t^{7-r} (t + 1)^r,$$

for sufficiently large values of $m = 7t + r$.

32.7 The $(0, 1)$ Nonsquare Case: $n \equiv -1 \pmod 4$

The theory for D-optimal $(0, 1)$-designs is most developed for the cases where $n \equiv -1 \pmod 4$.

Definitions:

For an $n \times n$ matrix A, the **trace-sequence** A is $(\text{trace}(A), \text{trace}(A^2), \cdots, \text{trace}(A^n))$.

$\mathcal{G}(v, \delta)$ is the set of all δ-regular graphs on v vertices.

Let graph G be a graph in $\mathcal{G}(v, \delta)$ and let $\mathcal{A}_G$ be the adjacency matrix of G. The graph G is **trace-minimal** if the **trace-sequence** of its adjacency matrix $(\text{trace}(\mathcal{A}_G), \text{trace}(\mathcal{A}_G^2), \cdots, \text{trace}(\mathcal{A}_G^n))$ is least in lexicographic order among all graphs in $\mathcal{G}(v, \delta)$.

Facts:

1. [AFN03] If $n \equiv -1 \pmod 4$, then for each $0 \leq r < n$ and all sufficiently large values of t, $\beta(nt + r, n)$ is a polynomial in t of degree n. These polynomials are related to the adjacency matrices $\mathcal{A}_G$ of certain regular graphs G.

2. [AFN03], [AFN06] The polynomial $\beta(nt + r, n)$ depends on a trace-minimal graph in $\mathcal{G}(v, \delta)$. Once a trace-minimal graph G is found in the appropriate graph class $\mathcal{G}(v, \delta)$, the polynomial $\beta(nt + r, n)$ can be computed. There are four theorems [AFN03] governing this situation; one for each congruence class of $r \pmod 4$.
3. [AFN03] Trace-minimal graphs are known for all of the graph classes necessary to obtain formulas for $\beta(nt + r, n)$ for $n = 3, 7, 11, 15, 19, 23,$ and 27 and t sufficiently large.
4. [AFN06] Let G be a connected strongly regular graph with no three cycles. Then G is trace-minimal.
5. [AFN06] The following graphs are trace-minimal in their graph class:

Graph Class	G
$\mathcal{G}(v, 0)$	Graph with v vertices and no edges
$\mathcal{G}(2v, 1)$	vK_2, a matching of $2v$ vertices
$\mathcal{G}(v, 2)$	C_v, the cycle graph on v vertices
$\mathcal{G}(2v, v)$	$K_{v,v}$, the complete bipartite graph with v vertices in each set of the bipartition
$\mathcal{G}(2v, 2v-2)$	$K_{2v} - vK_2$, the complement of a matching
$\mathcal{G}(v, v-1)$	K_v, the complete graph on v vertices

6. [AFN06] Let G be a connected regular graph with girth g such that $\mathcal{A}_G$ has $k+1$ distinct eigenvalues. If g is even, then $g \leq 2k$ with equality only if G is trace-minimal. If g is odd, then $g \leq 2k + 1$ with equality only if G is trace-minimal.

Examples:

1. Let $n = 4p - 1 \equiv -1 \pmod 4$ and $r = 4d + 2 \equiv 2 \pmod 4$. Let G be a trace-minimal graph in $\mathcal{G}(2p, p + d)$. Then
$$\beta(nt + r, n) = \frac{4t[p_{\mathcal{A}_G}(pt + d)]^2}{(t-1)^2},$$
for sufficiently large values of t. Taking $n = 15, r = 10$, we have $p = 4, d = 2$. The appropriate graph class is $\mathcal{G}(8, 6)$. There is only one graph G in this class, namely the complement of the matching $4K_2$. Thus, it is trace-minimal. Since $p_{\mathcal{A}_G}(x) = (x - 6)x^4(x + 2)^3$,
$$\beta(15t + 10, 15) = \frac{4t[p_{\mathcal{A}_G}(4t + 2)]^2}{(t-1)^2} = 16(4t)(4t + 2)^8(4t + 4)^6,$$
for sufficiently large t.
2. Let $n = 4p - 1 \equiv -1 \pmod 4$ and $r = 4d + 1 \equiv 1 \pmod 4$. Let G be a trace-minimal graph in $\mathcal{G}(2p, d)$. Then
$$\beta(nt + r, n) = \frac{4(t+1)[p_{\mathcal{A}_G}(pt + d)]^2}{t^2},$$
for sufficiently large t. Taking $n = 15, r = 9$ we have $p = 4, d = 2$. The appropriate graph class is $\mathcal{G}(8, 2)$. There are three (nonisomorphic) graphs in this class: C_8, $C_5 \cup C_3$, and $C_4 \cup C_4$, where C_k stands for a k-cycle graph. The trace sequences for these three graphs are
$$(\text{trace}(\mathcal{A}_G), \text{trace}(\mathcal{A}_G^2), \cdots, \text{trace}(\mathcal{A}_G^8)) = \begin{cases} (0, 16, 0, 48, 0, 160, 0, 576) \quad , & \text{for } G = C_8 \\ (0, 16, 6, 48, 40, 166, 196, 608), & \text{for } G = C_5 \cup C_3 \\ (0, 16, 0, 64, 0, 256, 0, 1024) \quad , & \text{for } G = C_4 \cup C_4. \end{cases}$$

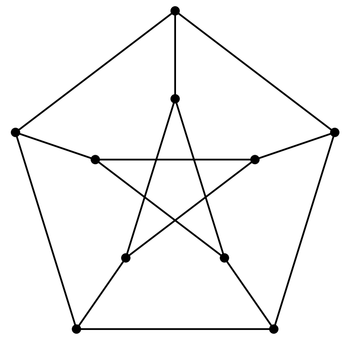

FIGURE 32.1

Thus, C_8 is the only trace-minimal graph in the graph class $\mathcal{G}(8,2)$. The characteristic polynomial for $\mathcal{A}_{C_8}$ is

$$(x-2)x^2(x+2)(x^2-2)^2.$$

Thus,

$$\beta(15t+9,15) = \frac{4(t+1)[p_{\mathcal{A}_G}(4t+2)]^2}{t^2} = 16(4t+2)^4(4t+4)^3(16t^2+16t+2)^4.$$

3. The Petersen graph (Figure 32.1) is an example of a strongly regular graph. (See Fact 4.) It is trace-minimal in $\mathcal{G}(10,3)$:
4. Let P be the projective geometry with seven points, $1,2,3,4,5,6,7$ and seven lines, $123, 147, 156, 257, 246, 367, 345$ (Figure 32.2): The line-point incidence matrix for P is:

$$N = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 \end{bmatrix}.$$

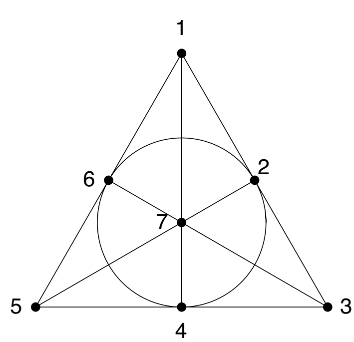

FIGURE 32.2

Let G be the incidence graph of P having 14 vertices and adjacency matrix given by

$$\mathcal{A}_G = \begin{bmatrix} 0 & N \\ N^T & 0 \end{bmatrix}.$$

Then G is trace-minimal by Fact 6: G is a regular graph of degree 3. The girth of G is $g = 6$. The characteristic polynomial of $\mathcal{A}_G$ is $(x-3)(x+3)(x^2-2)^6$ and so $\mathcal{A}_G$ has $k+1 = 4$ distinct eigenvalues. Since $2k = g$, it follows that G is trace-minimal in $\mathcal{G}(14, 3)$.

32.8 Balanced $(0, 1)$-Matrices and (± 1)-Matrices

Let $n = 2k - 1$ be odd. There is a connection between balanced $(0, 1)$-design matrices and (± 1)-design matrices.

Facts:

1. [NWZ98a] Let W be a balanced design matrix in $\{0, 1\}^{m\times n}$, so that each row of W contains exactly k ones and $k-1$ zeros. Let q be a positive integer and

$$L(W) = \begin{bmatrix} J_{n,1} & J_{n,m} - 2W^T \\ J_{q,1} & J_{q,m} \end{bmatrix}.$$

Then $\det L(W)^T L(W) = q4^m \det W^T W$. It follows that for sufficiently large m, if W is a balanced $(0, 1)$-design matrix and $L(W)$ is a D-optimal design matrix, then W is also D-optimal.

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33

Sign Pattern Matrices

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The origins of sign pattern matrices are in the book [Sam47] by the Nobel Economics Prize winner P. Samuelson, who pointed to the need to solve certain problems in economics and other areas based only on the signs of the entries of the matrices. The study of sign pattern matrices has become somewhat synonymous with qualitative matrix analysis. The dissertation of C. Eschenbach [Esc87], directed by C.R. Johnson, studied sign pattern matrices that "require" or "allow" certain properties and summarized the work on sign patterns up to that point. In 1995, Richard Brualdi and Bryan Shader produced a thorough treatment [BS95] on sign pattern matrices from the sign-solvability vantage point. There is such a wealth of information contained in [BS95] that it is not possible to represent all of it here. Since 1995 there has been a considerable number of papers on sign patterns and some generalized notions such as ray patterns. We remark that in this chapter we mostly use $\{+, -, 0\}$ notation for sign patterns, whereas in the literature $\{1, -1, 0\}$ notation is also commonly used, such as in [BS95]. We further note that because of the interplay between sign pattern matrices and graph theory, the study of sign patterns is regarded as a part of combinatorial matrix theory.

33.1 Basic Concepts

Definitions:

A **sign pattern matrix** (or **sign pattern**) is a matrix whose entries come from the set $\{+, -, 0\}$. For a real matrix B, $\text{sgn}(B)$ is the sign pattern whose entries are the signs of the corresponding entries in B.

If A is an $m \times n$ sign pattern matrix, the **sign pattern class** (or **qualitative class**) of A, denoted $Q(A)$, is the set of all $m \times n$ real matrices B with $\text{sgn}(B) = A$. If C is a real matrix, its qualitative class is given by $Q(C) = Q(\text{sgn}(C))$.

A **generalized sign pattern** $\tilde{A}$ is a matrix whose entries are from the set $\{+, -, 0, \#\}$, where # indicates an ambiguous sum (the result of adding + with $-$). The qualitative class of $\tilde{A}$ is defined by allowing the # entries to be completely free. Two generalized sign patterns are **compatible** if there is a common real matrix in their qualitative classes.

A **subpattern** $\hat{A}$ of a sign pattern A is a sign pattern obtained by replacing some (possibly none) of the nonzero entries in A with 0; this fact is denoted by $\hat{A} \preceq A$.

A **diagonal pattern** is a square sign pattern all of whose off-diagonal entries are zero. Similarly, standard matrix terms such as "tridiagonal" and "upper triangular" can be applied to sign patterns having the required pattern of zero entries.

A **permutation pattern** is a square sign pattern matrix with entries 0 and +, where the entry + occurs precisely once in each row and in each column.

A **permutational similarity** of the (square) sign pattern A is a product of the form $P^T AP$, where P is a permutation pattern.

A **permutational equivalence** of the sign pattern A is a product of the form $P_1 AP_2$, where P_1 and P_2 are permutation patterns.

The **identity** pattern of order n, denoted I_n, is the $n \times n$ diagonal pattern with + diagonal entries.

A **signature pattern** is a diagonal sign pattern matrix, each of whose diagonal entries is $+$ or $-$.

A **signature similarity** of the (square) sign pattern A is a product of the form SAS, where S is a signature pattern.

If P is a property referring to a real matrix, then a sign pattern A **requires** P if every real matrix in $Q(A)$ has property P, or **allows** P if some real matrix in $Q(A)$ has property P.

The **digraph** of an $n \times n$ sign pattern $A = [a_{ij}]$, denoted $\Gamma(A)$, is the digraph with vertex set $\{1, 2, \ldots, n\}$, where (i, j) is an arc iff $a_{ij} \neq 0$. (See Chapter 29 for more information on digraphs.)

The **signed digraph** of an $n \times n$ sign pattern $A = [a_{ij}]$, denoted $D(A)$, is the digraph with vertex set $\{1, 2, \ldots, n\}$, where (i, j) is an arc (bearing a_{ij} as its sign) iff $a_{ij} \neq 0$.

If $A = [a_{ij}]$ is an $n \times n$ sign pattern matrix, then a **(simple) cycle** of **length** k (or a k-**cycle**) in A is a formal product of the form $\gamma = a_{i_1 i_2} a_{i_2 i_3} \ldots a_{i_k i_1}$, where each of the elements is nonzero and the index set $\{i_1, i_2, \ldots, i_k\}$ consists of distinct indices. The **sign** (positive or negative) of a simple cycle in a sign pattern A is the actual product of the entries in the cycle, following the obvious rules that multiplication is commutative and associative, and $(+)(+) = +$, $(+)(-) = -$.

A **composite cycle** γ in A is a product of simple cycles, say $\gamma = \gamma_1\gamma_2 \ldots \gamma_m$, where the index sets of the γ_i's are mutually disjoint. If the length of γ_i is l_i, then the length of γ is $\sum_{i=1}^{m} l_i$, and the **signature** of γ is $(-)^{\sum_{i=1}^{m}(l_i-1)}$. A cycle γ is **odd** (**even**) when the length of the simple or composite cycle γ is odd (even).

If $A = [a_{ij}]$ is an $n \times n$ sign pattern matrix, then a **path** of length k in A is a formal product of the form $a_{i_1 i_2} a_{i_2 i_3} \ldots a_{i_k i_{k+1}}$, where each of the elements is nonzero and the indices $i_1, i_2, \ldots, i_{k+1}$ are distinct.

Facts:

1. Simple cycles and paths in an $n \times n$ sign pattern matrix A correspond to simple cycles and paths in the digraph of A. In particular, the path $a_{i_1 i_2} a_{i_2 i_3} \ldots a_{i_k i_{k+1}}$ in A corresponds to the path $i_1 \to i_2 \to \ldots \to i_{k+1}$ in the digraph of A.
2. If A is an $n \times n$ sign pattern, then each nonzero term in $\det(A)$ is the product of the signature of a composite cycle γ of length n in A with the actual product of the entries in γ.
3. Two generalized sign patterns are compatible if and only if the signs of each position whose sign is specified in both are equal.

Examples:

1. The matrix $\begin{bmatrix} 0 & 5 & -4 \\ -2 & -1 & 7 \end{bmatrix}$ is in $Q(A)$, where $A = \begin{bmatrix} 0 & + & - \\ - & - & + \end{bmatrix}$.

2. If $A = [a_{ij}] = \begin{bmatrix} + & + & 0 & - & + \\ 0 & - & - & - & + \\ + & - & - & + & + \\ - & - & + & - & + \\ + & - & 0 & - & - \end{bmatrix}$,

   then the composite cycle $\gamma = (a_{12}a_{23}a_{31})(a_{45}a_{54})$ has length 5 and negative signature, and yields the term $-a_{12}a_{23}a_{31}a_{45}a_{54} = -$ in $\det(A)$.
3. If $A = \begin{bmatrix} + & + \\ + & - \end{bmatrix}$, then $A^2 = \begin{bmatrix} + & \# \\ \# & + \end{bmatrix}$, which is compatible with $\begin{bmatrix} + & - \\ 0 & \# \end{bmatrix}$.

33.2 Sign Nonsingularity

Definitions:

A square sign pattern A is **sign nonsingular** (SNS) if every matrix $B \in Q(A)$ is nonsingular.

A **strong sign nonsingular** sign pattern, abbreviated an S^{2}NS-pattern, is an SNS-pattern A such that the matrix B^{-1} is in the same sign pattern class for all $B \in Q(A)$.

A **self-inverse sign pattern** is an S^{2}NS-pattern A such that $B^{-1} \in Q(A)$ for every matrix $B \in Q(A)$.

A **maximal** sign nonsingular sign pattern matrix is an SNS-pattern A where no zero entry of A can be set nonzero so that the resulting pattern is SNS.

A **nearly sign nonsingular** (NSNS) sign pattern matrix is a square pattern A having at least two nonzero terms in the expansion of its determinant, with precisely one nonzero term having opposite sign to the others.

A square sign pattern A is **sign singular** if every matrix $B \in Q(A)$ is singular.

The **zero pattern** of a sign pattern A, denoted $|A|$, is the $(0,+)$-pattern obtained by replacing each $-$ entry in A by a $+$.

Since a sign pattern may be represented by any real matrix in its qualitative class, many concepts defined on sign patterns (such as SNS and S^{2}NS) may be applied to real matrices.

Facts:

Most of the following facts can be found in [BS95, Chaps. 1–4 and 6–8].

1. The $n \times n$ sign pattern A is sign nonsingular if and only if $\det(A) = +$ or $\det(A) = -$, that is, in the standard expansion of $\det(A)$ into $n!$ terms, there is at least one nonzero term, and all the nonzero terms have the same sign.
2. An $n \times n$ pattern A is an SNS-pattern iff for any $n \times n$ signature pattern D and any $n \times n$ permutation patterns P_1, P_2, DP_1AP_2 is an SNS-pattern.
3. [BMQ68] For any SNS-pattern A, there exist a signature pattern D and a permutation pattern P such that DPA has negative diagonal entries.
4. [BMQ68] An $n \times n$ sign pattern A with negative main diagonal is SNS iff the actual product of entries of every simple cycle in A is negative.
5. [Gib71] If an $n \times n$, $n \geq 3$, sign pattern A is SNS, then A has at least $\binom{n-1}{2}$ zero entries, with exactly this number iff there exist permutation patterns P_1 and P_2 such that P_1AP_2 has the same zero/nonzero pattern as the Hessenberg pattern given in Example 1 below.
6. The fully indecomposable maximal SNS-patterns of order ≤ 9 are given in [LMV96]. [GOD96] An $n \times n$ sign pattern A is a fully indecomposable maximal SNS-pattern with $\binom{n-1}{2}$ zero entries iff A is equivalent (namely, one can be be transformed into the other by any combination of transposition, multiplication by permutation patterns, and multiplication by signature patterns) to the pattern given in Example 1 below. For $n \geq 5$, there is precisely one equivalence class of fully indecomposable

maximal SNS-patterns with $\binom{n-1}{2}+1$ zero entries, and there are precisely two such equivalence classes having $\binom{n-1}{2}+2$ zero entries.

7. [BS95, Corollary 1.2.8] If A is an $n \times n$ sign pattern, then A is an S^{2}NS-pattern iff

   (a) A is an SNS-pattern, and

   (b) For each i and j with $a_{ij}=0$, the submatrix $A(i,j)$ of A of order $n-1$ obtained by deleting row i and column j is either an SNS-pattern or a sign singular pattern.

8. [BMQ68] If A is an $n \times n$ sign pattern with negative main diagonal, then A is an S^{2}NS-pattern iff

   (a) The actual product of entries of every simple cycle in A is negative, and

   (b) The actual product of entries of every simple path in A is the same, for any paths with the same initial row index and the same terminal column index.

9. [LLM95] An irreducible sign pattern A is NSNS iff there exists a permutation pattern P and a signature pattern S such that $B = APS$ satisfies:

   (a) $b_{ii} < 0$ for $i = 1, 2, \ldots, n$.

   (b) The actual product of entries of every cycle of length at least 2 of $D(B)$ is positive.

   (c) $D(B)$ is intercyclic (namely, any two cycles of lengths at least two have a common vertex).

10. [Bru88] An $n \times n$ sign pattern A is SNS iff $\mathrm{per}(|B|) = |\det(B)|$, where B is the $(1,-1,0)$-matrix in $Q(A)$ and $|B|$ is obtained from B by replacing every -1 entry with 1.
11. [Tho86] The problem of determining whether a given sign pattern A is an SNS-pattern is equivalent to the problem of determining whether a certain digraph related to $D(A)$ has an even cycle.

For further reading, see [Kas63], [Bru88], [BS91], [BC92], [EHJ93], [BCS94a], [LMO96], [SS01], and [SS02].

Examples:

1. For $n \geq 2$, the following Hessenberg pattern is SNS:

$$H_n = \begin{bmatrix} - & + & 0 & \ldots & 0 & 0 \\ - & - & + & \ldots & 0 & 0 \\ - & - & - & \ldots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ - & - & - & \ldots & - & + \\ - & - & - & \ldots & - & - \end{bmatrix}.$$

2. [BS95, p. 11] For $n \geq 2$, the following Hessenberg pattern is S^{2}NS:

$$G_n = \begin{bmatrix} + & - & 0 & \ldots & 0 & 0 \\ 0 & + & - & \ldots & 0 & 0 \\ 0 & 0 & + & \ldots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \ldots & + & - \\ + & 0 & 0 & \ldots & 0 & + \end{bmatrix}.$$

3.

$$A = \begin{bmatrix} - & + & 0 & 0 \\ 0 & - & + & 0 \\ 0 & 0 & - & + \\ - & 0 & 0 & - \end{bmatrix}$$

is S^2NS with inverse pattern

$$\begin{bmatrix} - & - & - & - \\ + & - & - & - \\ + & + & - & - \\ + & + & + & - \end{bmatrix}.$$

4. [BS95, p. 114] The following patterns are maximal SNS-patterns:

$$\begin{bmatrix} - & + & 0 \\ - & - & + \\ - & - & - \end{bmatrix}, \quad \begin{bmatrix} - & + & 0 & + \\ - & - & + & 0 \\ 0 & - & - & + \\ - & 0 & - & - \end{bmatrix}.$$

33.3 Sign-Solvability, L-Matrices, and S^*-Matrices

Definitions:

A system of linear equations $Ax = b$ (where A and b are both sign patterns or both real matrices) is **sign-solvable** if for each $\tilde{A} \in Q(A)$ and for each $\tilde{b} \in Q(b)$, the system $\tilde{A}x = \tilde{b}$ is consistent and

$$\{\tilde{x} \ : \ \text{there exist } \tilde{A} \in Q(A) \text{ and } \tilde{b} \in Q(b) \text{ with } \tilde{A}\tilde{x} = \tilde{b}\}$$

is entirely contained in one qualitative class.

A sign pattern or real matrix A is an L**-matrix** if for every $B \in Q(A)$, the rows of B are linearly independent.

A **barely** L**-matrix** is an L-matrix that is not an L-matrix if any column is deleted.

An $n \times (n+1)$ matrix B is an S^***-matrix** provided that each of the $n+1$ matrices obtained by deleting a column of B is an SNS matrix.

An $n \times (n+1)$ matrix B is an S**-matrix** if it is an S^*-matrix and the kernel of every matrix in $Q(B)$ contains a vector all of whose coordinates are positive.

A **signing** of order k is a nonzero $(0, 1, -1)$- or $(0, +, -)$-diagonal matrix of order k.

A signing $D' = \mathrm{diag}(d_1', d_2', \ldots, d_k')$ is an **extension** of the signing D if $D' \neq D$ and $d_i' = d_i$ whenever $d_i \neq 0$.

A **strict signing** is a signing where all the diagonal entries are nonzero.

Let $D = \mathrm{diag}(d_1, d_2, \ldots, d_k)$ be a signing of order k and let A be an $m \times n$ (real or sign pattern) matrix. If $k = m$, then DA is a **row signing** of the matrix A, and if D is strict, then DA is a **strict row signing** of the matrix A. If $k = n$, then AD is a **column signing** of the matrix A, and if D is strict, then AD is a **strict column signing** of the matrix A.

A real or sign pattern vector is **balanced** provided either it is a zero vector or it has both a positive entry and a negative entry. A vector v is **unsigned** if it is not balanced. A **balanced row signing** of the matrix A is a row signing of A in which all the columns are balanced. A **balanced column signing** of the matrix A is a column signing of A in which all the rows are balanced.

Facts:

Most of the following facts can be found in [BS95, Chaps. 1–3].

1. A square sign pattern A is an L-matrix iff A is an SNS matrix.
2. The linear system $Ax = 0$ is sign-solvable iff A^T is an L-matrix
3. If $Ax = b$ is sign-solvable, then A^T is an L-matrix.
4. $Ax = b$ is sign-solvable for all b if A is a square matrix and there exists a permutation matrix P such that PA is an invertible diagonal matrix.
5. Sign-solvability has been studied using signed digraphs; see [Man82], [Han83], [BS95, Chap. 3], and [Sha00].
6. [BS95] Let A be a matrix of order n and let b be an $n \times 1$ vector. Then $Ax = b$ is sign-solvable iff A is an SNS-matrix and for each i, $1 \leq i \leq n$, the matrix $A(i \leftarrow b)$, obtained from A by replacing the i-th column by b, is either an SNS matrix or has an identically zero determinant.
7. [BS95] If $AX = B$ is a sign-solvable linear system where A and B are square matrices of order n and B does not have an identically zero determinant, then A is an S^{2}NS matrix.
8. [BS95] Let $Ax = b$ be a linear system such that A has no zero rows. Then the linear system $Ax = b$ is sign-solvable and the vectors in its qualitative solution class have no zero coordinates iff the matrix $[A \mid -b]$ is an S^*-matrix.
9. An $n \times (n+1)$ matrix B is an S^*-matrix iff there exists a vector w with no zero coordinates such that the kernels of the matrices $\tilde{B} \in Q(B)$ are contained in $\{0\} \cup Q(w) \cup Q(-w)$.
10. [Man82] and [KLM84] Let $A = [a_{ij}]$ be an $m \times n$ matrix and let b be an $m \times 1$ vector. Assume that $z = (z_1, z_2, \ldots, z_n)^T$ is a solution of the linear system $Ax = b$. Let

$$\beta = \{j \ : \ z_j \neq 0\} \text{ and } \alpha = \{i \ : \ a_{ij} \neq 0 \text{ for some } j \in \beta\}.$$

Then $Ax = b$ is sign-solvable iff the matrix $[A[\alpha, \beta] \mid -b[\alpha]]$ is an S^*-matrix and the matrix $A(\alpha, \beta)^T$ is an L-matrix.
11. [KLM84] A matrix A is an L-matrix iff every row signing of A contains a unisigned column.
12. [BCS94a] An $m \times n$ matrix A is a barely L-matrix iff
    (a) A is an L-matrix.
    (b) For each $i = 1, 2, \ldots, n$, there is a row signing of A such that column i is the only unisigned column.
13. [BCS94a] An $m \times n$ matrix A is an S^*-matrix iff $n = m + 1$ and every row signing of A contains at least two unisigned columns.
14. [BCS94a] An $m \times n$ matrix A is an S^*-matrix iff $n = m + 1$ and there exists a strict signing D such that AD and $A(-D)$ are the only balanced column signings of A.
15. [KLM84] A matrix A is an S-matrix iff A is an S^*-matrix and every row of A is balanced.
16. Let A be an $m \times n$ sign pattern that does not have a $p \times q$ zero submatrix for any positive integers p and q with $p + q \geq m$. Then A is an L-matrix iff every strict row signing of A has a unisigned column.

For further reading, see [Sha95], [Sha99], [KS00], and [SR04].

Examples:

1. $\begin{bmatrix} + & - & + & + \\ + & + & - & + \\ + & + & + & - \end{bmatrix}$ is an L-matrix by Fact 12, and is a barely L-matrix by Fact 5 of Section 33.2.

2. [BS95, p. 65] The $m \times (m+1)$ matrix

$$H'_{m+1} = \begin{bmatrix} - & + & 0 & \dots & 0 & 0 & 0 \\ - & - & + & \dots & 0 & 0 & 0 \\ - & - & - & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ - & - & - & \dots & - & + & 0 \\ - & - & - & \dots & - & - & + \end{bmatrix}$$

is an S-matrix. Every strict column signing $H'_{m+1}D$ of H'_{m+1} is an S^*-matrix, with only two such strict column signings yielding S-matrices (namely, when $D = \pm I$).

Applications:

[BS95, Sec. 1.1] In supply and demand analysis in economics, linear systems, where the coefficients as well as the constants have prescribed signs, arise naturally. For instance, the sign-solvable linear system

$$\begin{bmatrix} + & - \\ - & - \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ - \end{bmatrix}$$

arises from the study of a market with one product, where the price and quantity are determined by the intersection of its supply and demand curves.

33.4 Stability

Definitions:

A **negative stable** (respectively, **negative semistable**) real matrix is a square matrix B where each of the eigenvalues of B has negative (respectively, nonpositive) real part. In this section the term *(semi)stable* will mean *negative (semi)stable*. More information on matrix stability can be found in Section 9.5 and Chapter 19.

A **sign stable** (respectively, **sign semistable**) sign pattern matrix is a square sign pattern A where every matrix $B \in Q(A)$ is stable (respectively, semistable).

A **potentially stable** sign pattern matrix is a square sign pattern A where some matrix $B \in Q(A)$ is stable.

An $n \times n$ sign pattern matrix A allows a **properly signed nest** if there exists $B \in Q(A)$ and a permutation matrix P such that $\operatorname{sgn}(\det(P^T B P[\{1, \dots, k\}])) = (-1)^k$ for $k = 1, \dots, n$.

A **minimally potentially stable** sign pattern matrix is a potentially stable, irreducible pattern such that replacing any nonzero entry by zero results in a pattern that is not potentially stable.

Facts:

Many of the following facts can be found in [BS95, Chap. 10].

1. A square sign pattern A is sign stable (respectively, sign semistable) iff each of the irreducible components of A is sign stable (respectively, sign semistable).
2. [QR65] If A is an $n \times n$ irreducible sign pattern, then A is sign semistable iff
   (a) A has nonpositive main diagonal entries.
   (b) If $i \neq j$, then $a_{ij}a_{ji} \leq 0$.
   (c) The digraph of A is a doubly directed tree.

3. If A is an $n \times n$ irreducible sign pattern, then A is sign stable iff
   (a) A has nonpositive main diagonal entries.
   (b) If $i \neq j$, then $a_{ij}a_{ji} \leq 0$.
   (c) The digraph of A is a doubly directed tree.
   (d) A does not have an identically zero determinant.
   (e) There does not exist a nonempty subset β of $[1, 2, \ldots, n]$ such that each diagonal element of $A[\beta]$ is zero, each row of $A[\beta]$ contains at least one nonzero entry, and no row of $A[\bar{\beta}, \beta]$ contains exactly one nonzero entry.

   The original version of this result was in terms of matchings and colorings in a graph ([JKD77, Theorem 2]); the restatement given here comes from [BS95, Theorem 10.2.2].
4. An efficient algorithm for determining whether a pattern is sign stable is given in [KD77], and the sign stable patterns have been characterized in finitely computable terms in [JKD87].
5. The characterization of the potentially stable patterns is a very difficult open question.
6. The potentially stable tree sign patterns (see Section 33.5) for dimensions less than five are given in [JS89].
7. If an $n \times n$ sign pattern A allows a properly signed nest, then A is potentially stable.
8. In [JMO97], sufficient conditions are determined for an $n \times n$ zero–nonzero pattern to allow a nested sequence of nonzero principal minors, and a method is given to sign a pattern that meets these conditions so that it allows a properly signed nest. It is also shown that if A is a tree sign pattern that has exactly one nonzero diagonal entry, then A is potentially stable iff A allows a properly signed nest.
9. In [LOD02], a measure of the relative distance to the unstable matrices for a stable matrix is defined and extended to a potentially stable sign pattern, and the minimally potentially stable patterns are studied.

Examples:

1. [BS95] The pattern $\begin{bmatrix} - & + \\ - & - \end{bmatrix}$ is sign stable, while the pattern $\begin{bmatrix} 0 & + \\ - & 0 \end{bmatrix}$ is sign semistable, but not sign stable.
2. [JMO97] The matrix $\begin{bmatrix} -1 & 1 & 0 \\ -1 & -1 & 1 \\ 0 & -3 & 1 \end{bmatrix}$ has a (leading) properly signed nest, so that the pattern $\begin{bmatrix} - & + & 0 \\ - & - & + \\ 0 & - & + \end{bmatrix}$ is potentially stable.
3. [JMO97] The matrix $B = \begin{bmatrix} -3 & 1 & 0 \\ 0 & 0 & 1 \\ 8 & -3 & 0 \end{bmatrix}$ has -1 as a triple eigenvalue, and so is stable. Thus, the sign pattern $A = \text{sgn}(B)$ is potentially stable, but it does not have a properly signed nest.
4. [LOD02] The $n \times n$ tridiagonal sign pattern A with $a_{11} = -, a_{i,i+1} = +, a_{i+1,i} = -$ for $i = 1, \ldots, n-1$, and all other entries 0, is minimally potentially stable.

Applications:

[BS95, sec. 10.1] The theory of sign stability is very important in population biology ([Log92]). For instance, a general ecosystem consisting of n different populations can be modeled by a linear system of differential equations. The entries of the coefficient matrix of this linear system reflect the effects on the ecosystem due to a small perturbation. The signs of the entries of the coefficient matrix can often be determined from general principles of ecology, while the actual magnitudes are difficult to determine and can only be approximated. The sign stability of the coefficient matrix determines the stability of an equilibrium state of the system.

33.5 Other Eigenvalue Characterizations or Allowing Properties

Definitions:

A **bipartite sign pattern matrix** is a sign pattern matrix whose digraph is bipartite.

A **combinatorially symmetric sign pattern matrix** is a square sign pattern A, where $a_{ij} \neq 0$ iff $a_{ji} \neq 0$.

The **graph of a combinatorially symmetric** $n \times n$ **sign pattern matrix** $A = [a_{ij}]$ is the graph with vertex set $\{1, 2, \ldots, n\}$, where $\{i, j\}$ is an edge iff $a_{ij} \neq 0$.

A **tree sign pattern** (t.s.p.) matrix is a combinatorially symmetric sign pattern matrix whose graph is a tree (possibly with loops).

An n**-cycle pattern** is a sign pattern A where the digraph of A is an n-cycle.

A k**-consistent sign pattern matrix** is a sign pattern A where every matrix $B \in Q(A)$ has exactly k real eigenvalues.

Facts:

1. [EJ91] An $n \times n$ sign pattern A requires all real eigenvalues iff each irreducible component of A is a symmetric t.s.p. matrix.
2. [EJ91] An $n \times n$ sign pattern A requires all nonreal eigenvalues iff each irreducible component of A
   (a) Is bipartite.
   (b) Has all negative simple cycles.
   (c) Is SNS.
3. [EJ93] For an $n \times n$ sign pattern A, the following statements are equivalent for each positive integer $k \geq 2$:
   (a) The minimum algebraic multiplicity of the eigenvalue 0 occurring among matrices in $Q(A)$ is k.
   (b) A requires an eigenvalue with algebraic multiplicity k, with k maximal.
   (c) The maximum composite cycle length in A is $n - k$.
4. If an $n \times n$ sign pattern A does not require an eigenvalue with algebraic multiplicity 2 (namely, if A allows n distinct eigenvalues), then A allows diagonalizability.
5. Sign patterns that require all distinct eigenvalues have many nice properties, such as requiring diagonalizability. In [LH02], a number of sufficient and/or necessary conditions for a sign pattern to require all distinct eigenvalues are established. Characterization of patterns that require diagonalizability is still open.
6. [EHL94a] If a sign pattern A requires all distinct eigenvalues, then A is k-consistent for some k.
7. [EHL94a] Let A be an $n \times n$ sign pattern and let A_I denote the sign pattern obtained from A by replacing all the diagonal entries by $+$. Then A_I requires n distinct real eigenvalues iff A is permutation similar to a symmetric irreducible tridiagonal sign pattern.
8. [LHZ97] A 3×3 nonnegative irreducible nonsymmetric sign pattern A allows normality iff $AA^T = A^T A$ and A is not permutation similar to $\begin{bmatrix} + & + & 0 \\ + & 0 & + \\ + & + & + \end{bmatrix}$.

9. [LHZ97] If $\begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix}$ allows normality, where A_1 is square, then the square pattern

$$\begin{bmatrix} A_1 & A_2 & A_2 & \dots & A_2 \\ A_3 & A_4 & A_4 & \dots & A_4 \\ A_3 & A_4 & A_4 & \dots & A_4 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ A_3 & A_4 & A_4 & \dots & A_4 \end{bmatrix}$$

   also allows normality. Parallel results hold for allowing idempotence and for allowing nilpotence of index 2. (See [HL99] and [EL99].)
10. [EL99] Suppose an $n \times n$ sign pattern A allows nilpotence. Then A^n is compatible with 0, and for $1 \le k \le n-1$, $\text{tr}(A^k)$ is compatible with 0. Further, for each $m, 1 \le m \le n$, $E_m(A)$ (the sum of all principal minors of order m) is compatible with 0.
11. [HL99] Let A be a 5×5 irreducible symmetric sign pattern such that A^2 is compatible with A. Then A allows a symmetric idempotent matrix unless A can be obtained from the following by using permutation similarity and signature similarity:

$$\begin{bmatrix} + & + & + & + & 0 \\ + & + & - & 0 & - \\ + & - & + & + & + \\ + & 0 & + & + & - \\ 0 & - & + & - & + \end{bmatrix}.$$

12. [SG03] Let A be an $n \times n$ sign pattern. If the maximum composite cycle length in A is equal to the maximum rank (see section 33.6) of A, then A allows diagonalizability.
13. [SG03] Every combinatorially symmetric sign pattern allows diagonalizability.
14. A nonzero $n \times n$ $(n \ge 2)$ sign pattern that requires nilpotence does not allow diagonalizability.
15. Complete characterization of sign patterns that allow diagonalizability is still open.

For further reading, see [Esc93a], [Yeh96], and [KMT96].

Examples:

1. By Fact 1, the pattern

$$\begin{bmatrix} * & + & 0 & 0 \\ + & * & - & - \\ 0 & - & * & 0 \\ 0 & - & 0 & * \end{bmatrix},$$

   where each $*$ entry could be $0, +,$ or $-$, requires all real eigenvalues.
2. [LH02] Up to equivalence (negation, transposition, permutational similarity, and signature similarity), the 3×3 irreducible sign patterns that require 3 distinct eigenvalues are the irreducible tridiagonal symmetric sign patterns, irreducible tridiagonal skew-symmetric sign patterns, and 3-cycle sign patterns, together with the following:

$$\begin{bmatrix} + & + & 0 \\ 0 & 0 & + \\ + & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & + & 0 \\ - & 0 & + \\ + & - & 0 \end{bmatrix}, \begin{bmatrix} 0 & + & 0 \\ - & 0 & + \\ + & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & + & - \\ - & 0 & + \\ + & - & 0 \end{bmatrix}, \begin{bmatrix} + & + & 0 \\ 0 & 0 & + \\ + & - & 0 \end{bmatrix}.$$

3. [EL99] The sign pattern $A = \begin{bmatrix} + & + & 0 \\ - & - & - \\ - & + & + \end{bmatrix}$ does not allow nilpotence, though it satisfies many "obvious" necessary conditions.
4. [LHZ97] The $n \times n$ sign pattern $\begin{bmatrix} + & + & + & \cdots & + \\ + & + & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ + & + & 0 & \cdots & 0 \end{bmatrix}$ allows normality.

33.6 Inertia, Minimum Rank

Definitions:

Let A be a sign pattern matrix.

The **minimal rank** of A, denoted $\mathrm{mr}(A)$, is defined by $\mathrm{mr}(A) = \min\{ \text{ rank } B : B \in Q(A) \}$.

The **maximal rank** of A, $\mathrm{MR}(A)$, is given by $\mathrm{MR}(A) = \max\{ \text{ rank } B : B \in Q(A) \}$.

The **term rank** of A is the maximum number of nonzero entries of A no two of which are in the same row or same column.

For a symmetric sign pattern A, $\mathrm{smr}(A)$, the **symmetric minimal rank** of A is $\mathrm{smr}(A) = \min\{ \text{ rank } B : B = B^T, \ B \in Q(A) \}$.

The **symmetric maximal rank** of A, $\mathrm{SMR}(A)$, is $\mathrm{SMR}(A) = \max\{ \text{ rank } B : B = B^T, \ B \in Q(A) \}$.

For a symmetric sign pattern A, the **(symmetric) inertia set of** A is in $(A) = \{ \text{ in}(B) : B = B^T \in Q(A) \}$.

A **requires unique inertia** if $\mathrm{in}(B_1) = \mathrm{in}(B_2)$ for all symmetric matrices $B_1, B_2 \in Q(A)$.

A sign pattern A of order n is an **inertially arbitrary pattern** (IAP) if every possible ordered triple (p, q, z) of nonnegative integers $p, q,$ and z with $p + q + z = n$ can be achieved as the inertia of some $B \in Q(A)$.

A **spectrally arbitrary pattern** (SAP) is a sign pattern A of order n such that every monic real polynomial of degree n can be achieved as the characteristic polynomial of some matrix $B \in Q(A)$.

Facts:

1. $\mathrm{MR}(A)$ is equal to the term rank of A.
2. Starting with a real matrix whose rank is $\mathrm{mr}(A)$ and changing one entry at a time to eventually reach a real matrix whose rank is $\mathrm{MR}(A)$, all ranks between $\mathrm{mr}(A)$ and $\mathrm{MR}(A)$ are achieved by real matrices.
3. [HLW01] For every symmetric sign pattern A, $\mathrm{MR}(A)$=$\mathrm{SMR}(A)$.
4. [HS93] A sign pattern A requires a fixed rank r iff A is permutationally equivalent to a sign pattern of the form $\begin{bmatrix} X & Y \\ Z & 0 \end{bmatrix}$, where X is $k \times (r - k), 0 \leq k \leq r$, and Y and Z^T are L-matrices.
5. [HLW01] A symmetric sign pattern A requires a unique inertia iff $\mathrm{smr}(A)$=$\mathrm{SMR}(A)$.
6. [HLW01] For the symmetric sign pattern $A = \begin{bmatrix} 0 & A_1 \\ A_1^T & 0 \end{bmatrix}$ of order n, we have

$$\mathrm{in}(A) = \{(k, k, n - 2k) : \mathrm{mr}(A_1) \leq k \leq \mathrm{MR}(A_1)\}.$$

In particular, $2\ \mathrm{mr}(A_1) = \mathrm{smr}(A)$.
7. [DJO00], [EOD03] Let T_n be the $n \times n$ tridiagonal sign pattern with each superdiagonal entry positive, each subdiagonal entry negative, the $(1, 1)$ entry negative, the (n, n) entry positive, and every other entry zero. It is conjectured that T_n is an SAP for all $n \geq 2$. It is shown that for $3 \leq n \leq 16$, T_n is an SAP.

8. [GS01] Let S_n be the $n \times n\,(n \geq 2)$ sign pattern with each strictly upper (resp., lower) triangular entry positive (resp., negative), the $(1,1)$ entry negative, the (n,n) entry positive, and all other diagonal entries zero. Then S_n is inertially arbitrary.
   [ML02] Further, if the $(1,n)$ and $(n,1)$ entries of S_n are replaced by zero, the resulting sign pattern is also an IAP.
9. [CV05] Not every inertially arbitrary sign pattern is spectrally arbitrary.
10. [MOT03] Suppose $1 \leq p \leq n-1$. Then every $n \times n$ sign pattern with p positive columns and $n-p$ negative columns is spectrally arbitrary.

For further reading see [CHL03], [SSG04], [Hog05], and references [Nyl96], [JD99], [BFH04], [BF04], and [BHL04] in Chapter 34.

Examples:

1. [HLW01] Let

$$A = \begin{bmatrix} + & 0 & + & + \\ 0 & + & + & + \\ + & + & - & 0 \\ + & + & 0 & - \end{bmatrix}.$$

   Then $\mathrm{in}(A) = (2,2,0)$, $\mathrm{smr}(A) = SMR(A) = 4$, but $3 = \mathrm{mr}(A) < \mathrm{smr}(A) = 4$.
2. [HL01] Let J_n be the $n \times n$ sign pattern with all entries equal to $+$. Then

$$\mathrm{in}(J_n) = \{(s,t,n-s-t) : s \geq 1,\ t \geq 0,\ s+t \leq n\}.$$

3. [Gao01], [GS03], [HL01] Let A be the $n \times n\ (n \geq 3)$ sign pattern all of whose diagonal entries are zero and all of whose off-diagonal entries are $+$. Then

$$\mathrm{in}(A) = \{(s,t,n-s-t) : s \geq 1,\ t \geq 2,\ s+t \leq n\}.$$

33.7 Patterns That Allow Certain Types of Inverses

Definitions:

A sign pattern matrix A is **nonnegative (positive)**, denoted $A \geq 0 (A > 0)$, if all of its entries are nonnegative (positive).

An **inverse nonnegative (inverse positive)** sign pattern matrix is a square sign pattern A that allows an entrywise nonnegative (positive) inverse.

Let both B and X be real matrices or nonnegative sign pattern matrices. Consider the following conditions.

(1) $BXB = B$.
(2) $XBX = X$.
(3) BX is symmetric.
(4) XB is symmetric.

For a real matrix B, there is a unique matrix X satisfying all four conditions above and it is called the **Moore–Penrose inverse** of B, and denoted by $B^\dagger$. More generally, let $B\{i,j,\ldots,l\}$ denote the set of matrices X satisfying conditions $(i),(j),\ldots,(l)$ from among conditions (1)–(4). A matrix $X \in B\{i,j,\ldots,l\}$ is called an $(i,j,\ldots,l)$**-inverse** of B. For example, if (1) holds, X is called a (1)-inverse of B; if (1) and (2) hold, X is called a (1, 2)-inverse of B, and so forth. See Section 5.7.

For a nonnegative sign pattern matrix B, if there is a nonnegative sign pattern X satisfying (1) to (4), then X is unique and it is called the Moore–Penrose inverse of B. An $(i, j, \ldots, l)$-inverse of B is defined similarly as in the preceding paragraph.

Facts:

The first three facts below are contained in [BS95, Chap. 9].

1. [JLR79] Let A be an $n \times n$ $(+, -)$-pattern, where $n \geq 2$. Then A is inverse positive iff A is not permutationally equivalent to a pattern of the form $\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$, where $A_{12} < 0, A_{21} > 0$, the blocks A_{11}, A_{22} are square or rectangular, and one (but not both) of A_{11}, A_{22} may be empty.
2. The above result in [JLR79] is generalized in [FG81] to $(+, -, 0)$-patterns, and additional equivalent conditions are established. Let e denote the column vector of all ones, J the $n \times n$ matrix all of whose entries equal 1, and A' the matrix obtained from A by replacing all negative entries with zeros. For an $n \times n$ fully indecomposable sign pattern A, the following are equivalent:
   (a) A is inverse positive.
   (b) A is not permutationally equivalent to a pattern of the form $\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$, where $A_{12} \leq 0, A_{21} \geq 0$, the blocks A_{11}, A_{22} are square or rectangular, and one (but not both) of A_{11}, A_{22} may be empty.
   (c) The pattern $\begin{bmatrix} 0 & A \\ -A^T & 0 \end{bmatrix}'$ is irreducible.
   (d) There exists $B \in Q(A)$ such that $Be = B^T e = 0$.
   (e) There exists a doubly stochastic matrix D such that $(D - \frac{1}{n} J) \in Q(A)$.
3. [Joh83] For an $n \times n$ fully indecomposable sign pattern A, the following are equivalent: A is inverse nonnegative; A is inverse positive; $-A$ is inverse nonnegative; $-A$ is inverse positive.
4. [EHL97] If an $n \times n$ sign pattern A allows B and B^{-1} to be in $Q(A)$, then
   (a) $\mathrm{MR}(A) = n$.
   (b) A^2 is compatible with I.
   (c) adj A is compatible with A and $\det(A)$ is compatible with $+$, or, adj A is compatible with $-A$ and $\det(A)$ is compatible with $-$, where adj A is the adjoint of A.
5. In [EHL94b], the class $\mathcal{G}$ of all square patterns A that allow $B, C \in Q(A)$ where $BCB = B$ is investigated; it is shown for nonnegative patterns that $\mathcal{G}$ coincides with the class of square patterns that allow $B \in Q(A)$ where $B^3 = B$.
6. [HLR04] An $m \times n$ nonnegative sign pattern A has a nonnegative $(1, 3)$-inverse (Moore–Penrose inverse) iff A allows a nonnegative $(1, 3)$-inverse (Moore–Penrose inverse).

For further reading, see [BF87], [BS95, Theorem 9.2.6], [SS01], and [SS02].

Examples:

1. By Facts 2 and 3, the sign pattern

$$A = \begin{bmatrix} + & 0 & + & + \\ 0 & + & + & + \\ - & - & - & 0 \\ - & - & 0 & - \end{bmatrix}$$

is not inverse nonnegative.

2. [EHL97] The sign pattern

$$A = \begin{bmatrix} + & + & + & + \\ 0 & + & + & + \\ 0 & + & - & - \\ 0 & - & + & + \end{bmatrix}$$

satisfies all the necessary conditions in Fact 4, but it does not allow an inverse pair B and B^{-1} in $Q(A)$.

33.8 Complex Sign Patterns and Ray Patterns

Definitions:

A **complex sign pattern matrix** is a matrix of the form $A = A_1 + iA_2$ for some $m \times n$ sign patterns A_1 and A_2, and the **sign pattern class** or **qualitative class** of A is

$$Q(A) = \{B_1 + iB_2 \ : \ B_1 \in Q(A_1) \ \text{and} \ B_2 \in Q(A_2)\}\ .$$

Many definitions for sign patterns, such as SNS, extend in the obvious way to complex sign patterns.

The **determinantal region** of a complex sign pattern A is the set

$$S_A = \{\det(B) \ : \ B \in Q(A)\}.$$

A **ray pattern** is a matrix each of whose entries is either 0 or a ray in the complex plane of the form $\{re^{i\theta} \ : \ r > 0\}$ (which is represented by $e^{i\theta}$). The **ray pattern class** of an $m \times n$ ray pattern A is

$$Q(A) = \{B = [b_{pq}] \in M_{m\times n}(\mathbb{C}) : b_{pq} = 0 \text{ iff } a_{pq} = 0, \text{ and otherwise } \arg b_{pq} = \arg a_{pq}\}.$$

For $\alpha < \beta$, the **open sector** from the ray $e^{i\alpha}$ to the ray $e^{i\beta}$ is the set of rays $\{re^{i\theta} \ : \ r > 0, \alpha < \theta < \beta\}$.

The **determinantal region** of a ray pattern A is the set

$$R_A = \{\det(B) \ : \ B \in Q(A)\}.$$

An $n \times n$ ray pattern A is **ray nonsingular** if the Hadamard product $X \circ A$ is nonsingular for every entrywise positive $n \times n$ matrix X.

A **cyclically real** ray pattern is a square ray pattern A where the actual products of every cycle in A is real.

Facts:

1. [EHL98], [SS05] For a complex sign pattern A, the boundaries of S_A are always on the axes on the complex plane.
2. [EHL98], [SS05] For a sign nonsingular complex sign pattern A, S_A is either entirely contained in an axis of the complex plane or is an open sector in the complex plane with boundary rays on the axes.
3. [SS05] For a complex sign pattern or ray pattern A, the region $S_A\backslash\{0\}$ (or $R_A\backslash\{0\}$) is an open set (in fact, a disjoint union of open sectors) in the complex plane, except in the cases that S_A (R_A) is entirely contained in a line through the origin.
4. The results of [MOT97], [LMS00], and [LMS04] show that there is an entrywise nonzero ray nonsingular ray pattern of order n if and only if $1 \leq n \leq 4$.
5. [EHL00] An irreducible ray pattern A is cyclically real iff A is diagonally similar to a real sign pattern. More generally, a ray pattern A is diagonally similar to a real sign pattern iff A and $A + A^*$ are both cyclically real.

Examples:

1. [EHL98] If $A = \begin{bmatrix} + & - \\ + & + \end{bmatrix} + i \begin{bmatrix} + & 0 \\ 0 & - \end{bmatrix}$, then A is sign nonsingular and S_A is the open sector from the ray $e^{-i\pi/2}$ to the ray $e^{i\pi/2}$.
2. [MOT97] The ray pattern $\begin{bmatrix} e^{i\pi/2} & + & + & + \\ + & e^{i\pi/2} & + & + \\ + & + & e^{i\pi/2} & + \\ + & + & + & e^{i\pi/2} \end{bmatrix}$ is ray nonsingular.
3. [EHL00] Let

$$A = \begin{bmatrix} 0 & e^{-i\theta_1} & 0 & -e^{-i\theta_1} \\ 0 & + & -e^{-i\theta_2} & 0 \\ e^{i(\theta_1+\theta_2)} & -e^{i\theta_2} & 0 & -e^{i\theta_2} \\ e^{i\theta_1} & - & 0 & - \end{bmatrix},$$

where θ_1 and θ_2 are arbitrary. Then A is cyclically real, and A is diagonally similar (via the diagonal ray pattern $S = \text{diag}(+, e^{i\theta_1}, e^{i(\theta_1+\theta_2)}, -e^{i\theta_1})$ to

$$\begin{bmatrix} 0 & + & 0 & + \\ 0 & + & - & 0 \\ + & - & 0 & + \\ - & + & 0 & - \end{bmatrix}.$$

33.9 Powers of Sign Patterns and Ray Patterns

Definitions:

Let J_n (or simply J) denote the all + sign (ray) pattern of order n.

A square sign pattern or ray pattern A is **powerful** if all the powers $A^1, A^2, A^3, \ldots$, are unambiguously defined, that is, no entry in any A^k is a sum involving two or more distinct rays. For a powerful pattern A, the smallest positive integers $l = l(A)$ and $p = p(A)$ such that $A^l = A^{l+p}$ are called the **base** and **period** of A, respectively.

A square sign pattern or ray pattern A is k-**potent** if k is the smallest positive integer such that $A = A^{k+1}$.

Facts:

1. [LHE94] An irreducible sign pattern A with index of imprimitivity h (see Section 29.7) is powerful iff all cycles of A with lengths odd multiples of h have the same sign and all cycles (if any) of A with lengths even multiples of h are positive (see [SS04]). A sign pattern A is powerful iff for every positive integer d and for every pair of matrices $B, C \in Q(A)$, $\text{sgn}(B^d) = \text{sgn}(C^d)$.
2. [LHE94] Let A be an irreducible powerful sign pattern, with index of imprimitivity h. Then the base and the period of A are given by $l(A) = l(|A|)$, $p(A) = h$ if A does not have any negative cycles, and $p(A) = 2h$ if A has a negative cycle.
3. [Esc93b] The only irreducible idempotent sign pattern of order $n \geq 2$ is the all + sign pattern.
4. [LHE94], [SEK99], [SBS02] Every k-potent irreducible sign or ray pattern matrix is powerful.
5. [EL97] The maximum number of $-$ entries in the square of a sign pattern of order n is $n^2 - 2$; the maximum number of $-$ entries in the square of a $(+, -)$ sign pattern of order n is $\lfloor n^2/2 \rfloor$.
6. [HL01b] Let A be an $n \times n$ ($n \geq 3$) sign pattern. If A^2 has only one entry that is not nonpositive, then A^2 has at most $n^2 - n$ negative entries.
7. [LHS02] Let A be an irreducible ray pattern. Then A is powerful iff A is diagonally similar to a subpattern of $e^{i\alpha} J$ for some $\alpha \in \mathbb{R}$, where J is the all + ray pattern.

8. [LHS05] Suppose that $A = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}$ is a powerful ray pattern, where A_{11} (resp., A_{22}) is irreducible with index of imprimitivity h_1 (resp., h_2) and $0 \neq A_{11} \preceq c_1 J_{n_1}, 0 \neq A_{22} \preceq c_2 J_{n_2}$. If $A_{12} \neq 0$, then $\left(\frac{c_2}{c_1}\right)^{\mathrm{lcm}(h_1,h_2)} = 1$.

For further reading, the structures of k-potent sign patterns or ray patterns are studied in [SEK99], [Stu99], [SBS02], [LG01], and [Stu03].

Examples:

1. [LHE94] The reducible sign pattern $A = \begin{bmatrix} 0 & + & + & + \\ 0 & + & 0 & + \\ 0 & 0 & - & - \\ 0 & 0 & 0 & 0 \end{bmatrix}$ satisfies $A^2 = \begin{bmatrix} 0 & + & - & \# \\ 0 & + & 0 & + \\ 0 & 0 & + & + \\ 0 & 0 & 0 & 0 \end{bmatrix}$ and $A^3 = A$. Thus, A is 2-potent and yet A is not powerful.
2. [SEK99] Let P_n be the $n \times n$ circulant permutation sign pattern with $(1, 2)$ entry equal to +. Let Q_n be the sign pattern obtained from P_n by replacing the + in the (n, 1) position with a $-$. Then P_n is n-potent and Q_n is $2n$-potent.
3. [SBS02] Suppose that $3|k$. Let $A = \omega \begin{bmatrix} 0 & J_{p\times q} & 0 \\ 0 & 0 & J_{q\times r} \\ J_{r\times p} & 0 & 0 \end{bmatrix}$, where $J_{m\times n}$ denotes the all ones $m \times n$ matrix and ω^3 is a primitive $k/3$-th root of unity. Then A is a k-potent ray pattern.

33.10 Orthogonality

Definitions:

A square sign pattern A is **potentially orthogonal (PO)** if A allows an orthogonal matrix.

A square sign pattern A that does not have a zero row or zero column is **sign potentially orthogonal** (**SPO**) if every pair of rows and every pair of columns allows orthogonality.

Two vectors $\mathbf{x} = [x_1, \ldots, x_n]$ and $\mathbf{y} = [y_1, \ldots, y_n]$ are **combinatorially orthogonal** if $|\{i : x_i y_i \neq 0\}| \neq 1$.

Facts:

1. Every PO sign pattern is SPO.
2. [BS94], [Wat96] For $n \leq 4$, every $n \times n$ SPO sign pattern is PO.
3. [Wat96] There is a 5×5 fully indecomposable SPO sign pattern that is not PO.
4. [JW98] There is a 6×6 $(+, -)$ sign pattern that is SPO but not PO.
5. [BBS93] Let A be an $n \times n$ fully indecomposable sign pattern whose rows are combinatorially orthogonal and whose columns are combinatorially orthogonal. Then A has at least $4(n-1)$ nonzero entries. This implies that a conjecture of Fiedler [Fie64], which says a fully indecomposable orthogonal matrix of order n has at least $4(n-1)$ nonzero entries, is true.
6. [CJL99] For $n \geq 2$, there is an $n \times n$ fully indecomposable orthogonal matrix with k zero entries iff $0 \leq k \leq (n-2)^2$.
7. [EHH99] Let S be any skew symmetric sign pattern of order n all of whose off-diagonal entries are nonzero. Then $I + S$ is PO.
8. [EHH99] It is an open question as to whether every sign pattern A that allows an inverse in $Q(A^T)$ is PO.

For further reading see [Lim93], [Sha98], [CS99], and [CHR03].

Examples:

1. [BS94] Every $3 \times 3 \pm$ SPO sign pattern can be obtained from the following sign pattern by using permutation equivalence and multiplication by signature patterns:

$$\begin{bmatrix} + & + & + \\ + & + & - \\ + & - & + \end{bmatrix}.$$

2. [Wat96] The sign pattern

$$\begin{bmatrix} - & + & 0 & + & - \\ + & + & - & 0 & - \\ 0 & + & + & + & + \\ + & 0 & - & + & + \\ - & - & - & + & + \end{bmatrix}$$

   is SPO but not PO.

33.11 Sign-Central Patterns

Definitions:

A real matrix B is **central** if the zero vector is in the convex hull of the columns of B. A real or sign pattern matrix A is **sign-central** if A requires centrality.

A **minimal sign-central matrix** A is a sign-central matrix that is not sign-central if any column of A is deleted.

A **tight sign-central** matrix is a sign-central matrix A for which the Hadamard (entrywise) product of any two columns of A contains a negative component.

A **nearly sign-central matrix** is a matrix that is not sign-central but can be augmented to a sign-central matrix by adjoining a column.

Facts:

1. [AB94] An $m \times n$ matrix A is sign-central iff the matrix DA has a nonnegative column vector for every strict signing D of order m.
2. [HKK03] Every tight sign-central matrix is a minimal sign-central matrix.
3. [LC00] If A is nearly sign-central and $[A \mid \alpha]$ is sign-central, then $[A \mid \alpha']$ is also sign-central for every $\alpha' \neq 0$ obtained from α by zeroing out some of its entries.

For further reading, see [BS95, Sect. 5.4], [DD90], [LLS97], and [BJS98].

Examples:

1. [BS95, p. 100], [HKK03] For each positive integer m, the $m \times 2^m \pm$ sign pattern E_m such that each m-tuple of $+$'s and $-$'s is a column of E_m, is a tight sign-central sign pattern.

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34

Multiplicity Lists for the Eigenvalues of Symmetric Matrices with a Given Graph

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This chapter assumes basic terminology from graph theory in Chapter 28; a good general graph theory reference is [CL96]. For standard terms or concepts from matrix analysis, see Part 1: Basic Linear Algebra, particularly Chapter 4.3, and Chapter 8; a good general matrix reference is [HJ85]. As we will be interested in properties of A that are permutation similarity invariant, primarily eigenvalues and their multiplicities, we will generally view a graph as unlabeled, except when referencing by labels is convenient.

For a given simple graph G on n vertices, let $\mathcal{S}(G)$ (respectively, $\mathcal{H}(G)$) denote the set of all $n\times n$ real symmetric (respectively, complex Hermitian) $n\times n$ matrices $A = [a_{ij}]$ such that for $i \neq j$, $a_{ij} \neq 0$ if and only if there is an edge between i and j.

Our primary interest lies in the following very general question. Given G, what are all the possible lists of multiplicities for the eigenvalues that occur among matrices in $\mathcal{S}(G)$ (respectively, $\mathcal{H}(G)$)? Much of our focus here is on the case in which $G = T$ is a tree.

It is important to distinguish two possible interpretations of "multiplicity list." Since the eigenvalues of a real symmetric or complex Hermitian matrix are real numbers, they may be placed in numerical order. If the multiplicities are placed in an order corresponding to the numerical order of the underlying eigenvalues, then we refer to such a way of listing the multiplicities as *ordered multiplicities*. If, alternatively, the multiplicities are simply listed in nonincreasing order of the values of the multiplicities themselves, we refer to such a list as *unordered multiplicities*. For example, if A has eigenvalues $-3, 0, 0, 1, 2, 2, 2, 5, 7$, the list of ordered multiplicities is $(1, 2, 1, 3, 1, 1)$, while the list of unordered multiplicities is $(3, 2, 1, 1, 1, 1)$. In either case, such a list means that there are exactly 6 different eigenvalues, of which 4 have multiplicity 1.

If a graph G is not connected, then the multiplicity lists for G may be deduced from those of its components via superposition. Also, graphs with many edges admit particularly rich collections of multiplicity lists. For example, the complete graph admits all multiplicity lists with the given number of eigenvalues, except the list in which all eigenvalues are the same. For these reasons, a natural beginning for the study of multiplicity lists for $\mathcal{S}(G)$ or $\mathcal{H}(G)$ is the case in which $G = T$, a tree. In addition, trees present several attractive features for this problem, so much of the research in this area, and this chapter, focuses on trees.

34.1 Multiplicities and Parter Vertices

Definitions:

Let G be a simple graph.

For an $n\times n$ real symmetric or complex Hermitian matrix $A = [a_{ij}]$, the **graph** of A, denoted by $G(A)$, is the simple graph on n vertices, labeled $1, 2, \ldots, n$, with an edge between i and j if and only if $a_{ij} \neq 0$.

Let $\mathcal{S}(G)$ (respectively, $\mathcal{H}(G)$), denote the set of all $n\times n$ real symmetric (respectively, complex Hermitian) matrices A such that $G(A) = G$ (where G has n vertices). No restriction is placed upon the diagonal entries of A by G, except that they are real.

If G' is the subgraph of G induced by β, $A(G')$ can be used to denote $A(\beta)$ and $A[G']$ to denote $A[\beta]$.

Given a tree T, the components of $T \setminus \{j\}$ are called **branches** of T at j.

If $A \in \mathcal{H}(G)$, $\lambda \in \sigma(A)$, and j is an index such that $\alpha_{A(j)}(\lambda) = \alpha_A(\lambda) + 1$, then j is called a **Parter index** or **Parter vertex** (for λ, A and G) (where $\alpha_A(\lambda)$ denotes the multiplicity of λ). Some authors refer to such a vertex as a **Parter–Wiener vertex** or a **Wiener vertex**.

If j is a Parter vertex for an eigenvalue λ of an $A \in \mathcal{H}(G)$ such that λ occurs as an eigenvalue of at least three direct summands of $A(j)$, j is called a **strong Parter vertex**.

A **downer vertex** i in a graph G (for $\lambda \in \sigma(A)$ and $A \in \mathcal{H}(G)$) is a vertex i such that $\alpha_{A(i)}(\lambda) = \alpha_A(\lambda) - 1$.

A **downer branch** of a tree T at j is a branch T_i at j, determined by a neighbor i of j such that i is a downer vertex in T_i (for λ and $A[T_i]$).

If a branch of a tree at a vertex j is a path and the neighbor of j in this branch is a pendant vertex of this path, the branch is a **pendant path** at j.

Facts:

1. If $A \in \mathcal{H}(G)$, then trivially $A(i) \in \mathcal{H}(G \setminus \{i\})$.
2. [HJ85] (Interlacing Inequalities) If an $n\times n$ Hermitian matrix A has eigenvalues $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ and $A(i)$ has eigenvalues $\beta_{i,1} \leq \beta_{i,2} \leq \cdots \leq \beta_{i,n-1}$, then $\lambda_1 \leq \beta_{i,1} \leq \lambda_2 \leq \beta_{i,2} \leq \cdots \leq \beta_{i,n-1} \leq \lambda_n$, $i = 1, \ldots, n$.
3. If λ is an eigenvalue of an Hermitian matrix A, then $\alpha_A(\lambda) - 1 \leq \alpha_{A(i)}(\lambda) \leq \alpha_A(\lambda) + 1$ for $i = 1, \ldots, n$.
4. If T is a tree, then any matrix of $\mathcal{H}(T)$ is diagonally unitarily similar to one in $\mathcal{S}(T)$.
5. [JLS03a](Parter–Wiener Theorem: Generalization) Let T be a tree and $A \in \mathcal{S}(T)$. Suppose that there exists an index i and a real number λ such that $\lambda \in \sigma(A) \cap \sigma(A(i))$. Then
   - There is in T a Parter vertex j for λ.
   - If $\alpha_A(\lambda) \geq 2$, then j may be chosen so that $\delta_T(j) \geq 3$ and so that there are at least three components T_1, T_2, and T_3 of $T \setminus \{j\}$ such that $\alpha_{A[T_k]}(\lambda) \geq 1$, $k = 1, 2, 3$.
   - If $\alpha_A(\lambda) = 1$, then j may be chosen so that there are two components T_1 and T_2 of $T\setminus\{j\}$ such that $\alpha_{A[T_k]}(\lambda) = 1$, $k = 1, 2$.
6. [JLS03a] For $A \in \mathcal{S}(T)$, T a tree, j is a Parter vertex for λ if and only if there is a downer branch at j for λ.

7. [JL] Suppose that G is a simple graph on n vertices that is not a tree. Then
   - There is a matrix $A \in \mathcal{S}(G)$ with an eigenvalue λ such that there is an index j so that $\alpha_A(\lambda) = \alpha_{A(j)}(\lambda) = 1$ and $\alpha_{A(i)}(\lambda) \leq 1$ for every $i = 1, \ldots, n$.
   - There is a matrix $B \in \mathcal{S}(G)$ with an eigenvalue λ such that $\alpha_B(\lambda) \geq 2$ and $\alpha_{B(i)}(\lambda) = \alpha_B(\lambda) - 1$, for every $i = 1, \ldots, n$.
8. [JLSSW03] Let T be a tree and $\lambda_1 < \lambda_2$ be eigenvalues of $A \in \mathcal{S}(T)$ that share a Parter vertex in T. Then there is at least one $\lambda \in \sigma(A)$ such that $\lambda_1 < \lambda < \lambda_2$.
9. Let T be a tree and $A \in \mathcal{S}(T)$. If T' is a pendant path of a vertex j of T, then T' is a downer branch for each eigenvalue of the direct summand $A[T']$.
10. Let T be a path and $A \in \mathcal{S}(T)$. Then each eigenvalue of A has multiplicity 1.
11. Let A be an $n \times n$ irreducible real symmetric tridiagonal matrix. Then
    - A has distinct eigenvalues.
    - In $A(i)$, there are at most $\min\{i-1, n-i\}$ interlacing equalities and this number may occur.
    - For each interlacing equality that does occur, the relevant eigenvalues must be an eigenvalue (of multiplicity 1) of both irreducible principal submatrices of $A(i)$.

    See Example 3 below.

Examples:

1. In general, one might expect that in passing from A to $A(i)$, multiplicities typically decline. However, Fact 5 is counter to this intuition in the case for trees. A rather complete statement has evolved through a series of papers ([Par60], [Wie84], [JLS03a]). In particular, Fact 5 says that when T is a tree and $\alpha_A(\lambda) \geq 2$, there must be a strong Parter vertex because, by interlacing, the hypothesis $\lambda \in \sigma(A) \cap \sigma(A(i))$ must be satisfied for any i. However, i itself need not be a Parter vertex. Even when $\alpha_A(\lambda) \geq 2$, it can happen that $\alpha_{A(j)}(\lambda) = \alpha_A(\lambda) + 1$ with $\delta_T(j) = 1$ or $\delta_T(j) = 2$ or λ appears in only one or two components of $T \setminus \{j\}$, even if $\delta_T(j) \geq 3$. There may, as well, be several Parter vertices and even several strong Parter vertices. Much information about Parter vertices may be found in [JLSSW03] and [JLS03a]. Let $\lambda, \mu \in \mathbb{R}$, $\lambda \neq \mu$, and consider real symmetric matrices whose graphs are the following trees, assuming that every diagonal entry corresponds to the label of the corresponding vertex.
   - The vertex v is a Parter vertex for λ in real symmetric matrices for which the graph is each of the trees in Figure 34.1. We also note that, depending on the tree T, several different vertices of T could be Parter for an eigenvalue of the same matrix in $\mathcal{S}(T)$. The matrices $A[T_1]$ and $A[T_2]$ each have u and v as Parter vertices for λ.

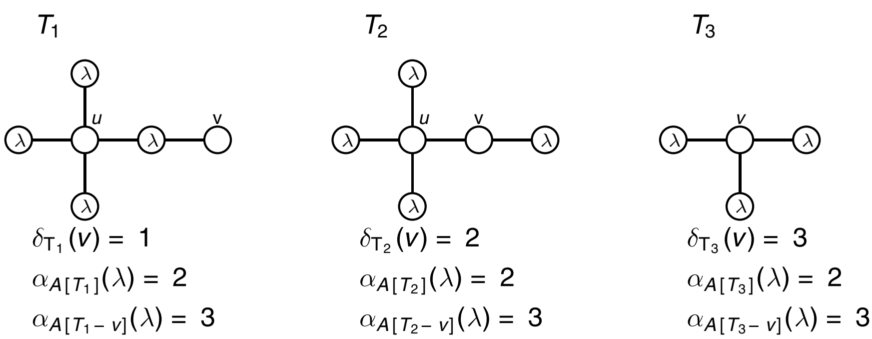

FIGURE 34.1 Examples of Parter vertices.

- Also, depending on the tree T, the same vertex could be a Parter vertex for different eigenvalues of a matrix in $\mathcal{S}(T)$. The vertex v is a Parter vertex for λ and μ in a real symmetric matrix A for which the graph is the tree in Figure 34.2. Such a matrix A has λ and μ as eigenvalues with $\alpha_A(\lambda) = 2 = \alpha_A(\mu)$. Since it is clear that we have $\alpha_{A(v)}(\lambda) = 3 = \alpha_{A(v)}(\mu)$, it means that v is Parter for λ and μ.

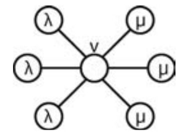

FIGURE 34.2 Vertex v is a Parter vertex for λ and μ.

2. Though a notion of "Parter vertex" can be defined for nontrees, Fact 7 is the converse to Fact 5 that shows that its remarkable conclusions are generally valid only for trees.
   Consider the matrix J_3 whose graph is the cycle C_3 (the possible multiplicities for the eigenvalues of a matrix whose graph is a cycle was studied in [Fer80]), which is not a tree. The matrix J_3 has eigenvalues $0, 0, 3$. Since the removal of any vertex from C_3 leaves a path, we conclude that there is no Parter vertex for the multiple eigenvalue 0.
3. If a graph G is a path on n vertices, then G is a tree and if the vertices are labeled consecutively, any matrix in $\mathcal{S}(G)$ is an irreducible tridiagonal matrix. Conversely, the graph of an irreducible real symmetric tridiagonal matrix is a path. The very special spectral structure of such matrices has been of interest for some time for a variety of reasons. Two well-known classical facts are that all eigenvalues are distinct (i.e., all 1s is the only multiplicity list) and, if a pendant vertex is deleted, the interlacing inequalities are strict. Both statements follow from Fact 5, but more can be gotten from Fact 5 as well. If A is $n \times n$ real symmetric and $1 \leq i \leq n$, then as many as $n - 1$ of the eigenvalues of $A(i)$ might coincide with some eigenvalue of A. We refer to such an occurrence as an "interlacing equality." If a pendant vertex is removed from a path, no interlacing equalities can occur, but if an interior vertex is removed, interlacing equalities can occur. The complete picture in this regard may be also be deduced from Fact 5.

34.2 Maximum Multiplicity and Minimum Rank

Definitions:

Let G be a simple graph.

The **maximum multiplicity** of G, $M(G)$, is the maximum multiplicity for a single eigenvalue among matrices in $\mathcal{S}(G)$.

The **minimum rank** of G is $\text{mr}(G) = \min_{A \in \mathcal{S}(G)} \text{rank}(A)$.

The **path cover number** of G, $P(G)$, is the minimum number of induced paths of G that do not intersect, but do cover all vertices of G.

$\Delta(T) = \max[p - q]$ over all ways in which q vertices may be deleted from G, so as to leave p paths. Isolated vertices count as (degenerate) paths.

The **maximum rank deficiency** for G is $m(G) = n - mr(G)$, where the order of G is n.

Facts:

Let G be a simple connected graph of order n.

1. If G is a path, $P(G) = 1$; otherwise $P(G) > 1$.
2. There may be many minimum path covers. See Example 2.
3. Maximizing sets of removed vertices (used in the computation of $\Delta(G)$) are not unique; even the number of removed vertices is not unique. See Example 3.

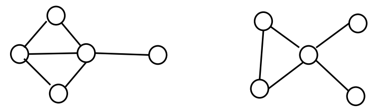

FIGURE 34.3 Two of the forbidden graphs for $\mathrm{mr}(G) = 2$.

4. [Fie69] $M(G) = 1$ if and only if G is a path.
5. $M(G) = n - 1$ if and only if G is the complete graph K_n.
6. $M(G) = m(G)$.
7. [JL99] For a tree T, $M(T) = \Delta(T) = P(T) = m(T)$.
8. [JS02] For any tree T, let H_T denote the subgraph of T induced by the vertices of degree at least 3. For a tree T, $\Delta(T)$ can be computed by the following algorithm. See Example 4.

Algorithm 1: Computation of $\Delta(T)$
Given a tree T.

1. Set $Q = \emptyset$ and $T' = T$.
2. While $H_{T'} \neq \emptyset$:
   Remove from T' all vertices v of $H_{T'}$ such that $\delta_{T'}(v) - \delta_{H_{T'}}(v) \geq 2$
   and add these vertices to Q.
3. $\Delta(T) = p - |Q|$ where p is the number of components (all of which are paths) in $T \backslash Q$.

9. [JL99], [BFH04] $\Delta(G) \leq M(G)$ and $\Delta(G) \leq P(G)$.
10. [BFH04], [BFH05] If $n \geq 2$, $P(K_n) < M(K_n)$, where K_n is the complete graph on n vertices. If G is unicyclic (i.e., has a unique cycle), then $P(G) \geq M(G)$ and strict inequality is possible.
11. [BFH04] Minimum rank (and, thus, maximum multiplicity) of a graph with a cut vertex can be computed from the minimum ranks of induced subgraphs.
12. [BHL04] If H is an induced subgraph of G, then $\mathrm{mr}(H) \leq \mathrm{mr}(G)$. Furthermore, $\mathrm{mr}(G) = 2$ (i.e., $M(G) = n-2$) if and only if G does not contain as an induced subgraph one of the following four forbidden graphs: the path on 4 vertices P_4, the complete tripartite graph $K_{3,3,3}$, the two graphs shown in Figure 34.3. Other characterizations are also given.

Examples:

1. Considering the tree T in Figure 34.4, we have $P(T) = 2$ (e.g., 1-3-2 and 5-4-6 constitute a minimal path cover of the vertices) and, of course, $\Delta(T) = 2$, as removal of vertex 4 leaves the 3 paths 1-3-2, 5, and 6 (and neither can be improved upon). Note that if submatrices $A[\{1, 2, 3\}]$, $A[\{5\}]$, and $A[\{6\}]$ of $A \in \mathcal{S}(T)$ are constructed so that λ is an eigenvalue of each (this is always possible and no higher multiplicity in any of them is possible), then $\alpha_A(\lambda) \geq 3 - 1 = 2$, which is the maximum possible.

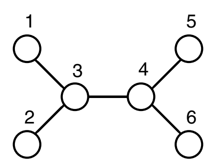

FIGURE 34.4 A tree with path cover number 2.

2. Consider the tree T on 12 vertices in Figure 34.5. It is not difficult to see that the path cover number of T, $P(T)$, is 4. However, it can be achieved by different collections of paths. For example, $P(T)$ can be achieved from the collection of 4 paths of T, 1-2-3, 4-5-6, 7-8 and 12-9-10-11. Similarly,

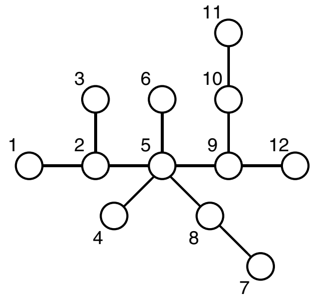

FIGURE 34.5 A tree with path cover number 4.

the paths of T, 1-2-3, 4-5-8-7, 6 and 12-9-10-11, form a collection of vertex disjoint paths (each one is an induced subgraph of T) that cover all the vertices of T.

3. Consider the tree T on 12 vertices in Figure 34.5. As we can see in Table 34.1, $\Delta(T)$ can be achieved for $q = 1, 2, 3$. When $q = 2$, there are 3 different sets of vertices whose removal from T leaves 6 components (paths), i.e., $p - q = 4$.
4. The algorithm in Fact 8 applied to the tree T in Figure 34.6 gives, in one step, a subset of vertices of T, $Q = \{v_2, v_3, v_4, v_5\}$, with cardinality 4 and such that $T \backslash Q$ has 13 components, each of which is a path. Therefore, $\Delta(T) = 13 - 4 = 9$.

   Note that, in any stage of the process to determine $\Delta(T)$, we may not choose just any vertex with degree greater than or equal to 3.

TABLE 34.1 $\Delta(T)$ for the tree T in Figure 34.5

Removed Vertices from T	q	p	$p - q$ $(= \Delta(T))$
5	1	5	4
5, 2	2	6	4
5, 9	2	6	4
5, 10	2	6	4
2, 5, 9	3	7	4
2, 5, 10	3	7	4

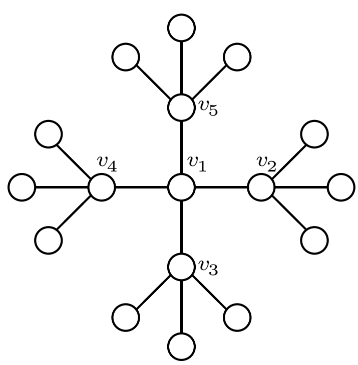

FIGURE 34.6 A tree T with $\Delta(T) = 9$.

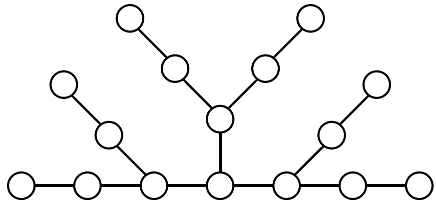

FIGURE 34.7 A tree of diameter 6 for which the minimum number of distinct eigenvalues is $8 > 6 + 1$.

34.3 The Minimum Number of Distinct Eigenvalues

Definitions:

Let T be a tree.

The **diameter** of T, $d(T)$, is the maximum number of edges in a path occurring as an induced subgraph of T.

The **minimum number of distinct eigenvalues** of T, $\mathcal{N}(T)$, is the minimum, among $A \in \mathcal{S}(T)$, number of distinct eigenvalues of A.

Facts:

1. [JL02a] Let T be a tree. Then $\mathcal{N}(T) \geq d(T) + 1$.
2. [JSa2] If T is a tree such that $d(T) < 5$, then there exist matrices in $\mathcal{S}(T)$ attaining as few distinct eigenvalues as $d(T) + 1$.

Examples:

1. Since each entry in a multiplicity list represents a distinct eigenvalue, the "length" of a list represents the number of different eigenvalues. This number can be as large as n (the number of vertices), of course, but it cannot be too small. Restrictions upon length limit the possible multiplicity lists. Just as a path has many distinct eigenvalues, a long (chordless) path occurring as an induced subgraph of a tree forces a large number of distinct eigenvalues.
2. For many trees T, there exist matrices in $\mathcal{S}(T)$ attaining as few distinct eigenvalues as $d(T) + 1$. However, for the tree T in Figure 34.7, $d(T) = 6$ and, in [BF04], the authors have shown that $\mathcal{N}(T) = 8 > d(T)+1$. It is not known how to deduce the minimum number of distinct eigenvalues from the structure of the tree, in general.

34.4 The Number of Eigenvalues Having Multiplicity 1

Definitions:

Given a tree T, let $\mathcal{U}(T)$ be the minimum number of 1s among multiplicity lists occurring for T.

Facts:

1. [JLS03a] For any tree T, the largest and smallest eigenvalues of any $A \in \mathcal{S}(T)$ necessarily have multiplicity 1.
2. [JLS03a] For any tree T on $n \geq 2$ vertices, $\mathcal{U}(T) \geq 2$ and, for each n, there exist trees T for which $\mathcal{U}(T) = 2$.
3. Let T be a tree on n vertices. $\mathcal{U}(T) \geq 2\mathcal{N}(T) - n$. In particular, $\mathcal{U}(T) \geq 2(d(T) + 1) - n$.